

Database Management System

Subject Code : 2114112

Module : 2

--Shilpa Ingoley

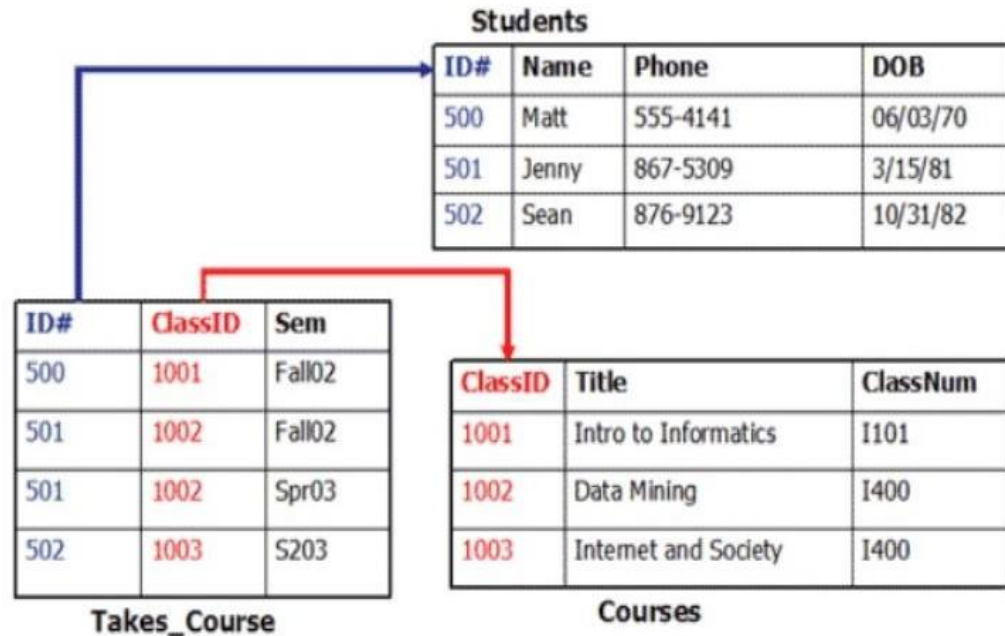
Module II: Relational Model and relational Algebra

- Introduction to the Relational Model, relational schema and concept of keys. Mapping the ER and EER Model to the Relational Model, Relational Algebra- operators
(Selection(σ), Projection(π), Union($\cup$), Difference ($-$), Cartesian Product($\times$), Join($\bowtie$), Intersection ($\cap$), Rename (ρ)), Relational Algebra Queries
- Self-learning Topics: Practice writing queries to perform common database tasks (e.g., selecting data, joining tables)

Introduction to the Relational Model, relational schema and concept of keys.

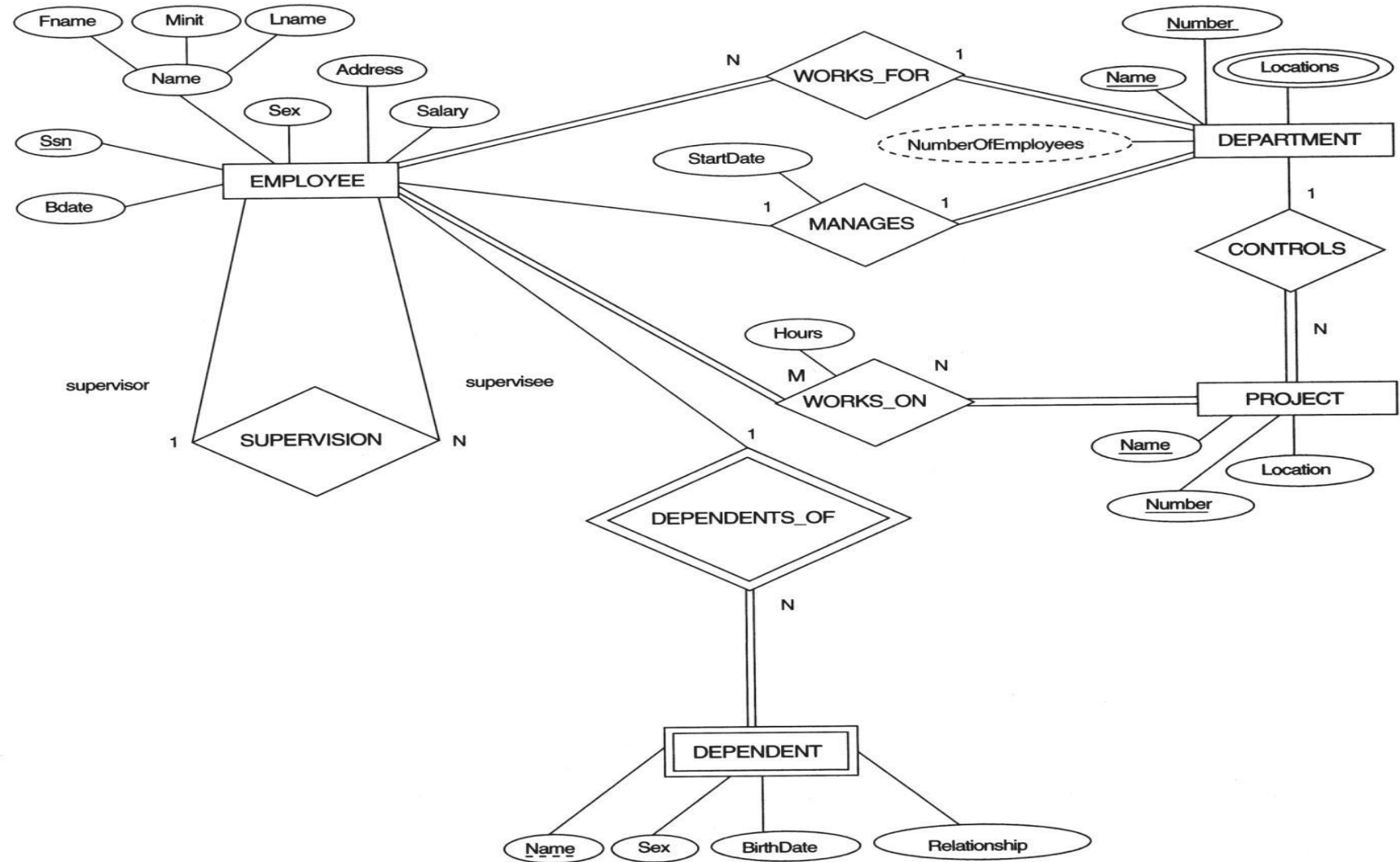
Relational database

Relational DBMS



- Stores data in the form of rows(tuple) and columns(attributes), and together forms a table(relation).
- It uses SQL for storing, manipulating, as well as maintaining the data.

Mapping the ER and EER Model to the Relational Model



Steps for mapping

- **ER-to-Relational Mapping Algorithm**

Step 1: Mapping of Regular Entity Types

Step 2: Mapping of Weak Entity Types

Step 3: Mapping of Binary 1:1 Relation Types

Step 4: Mapping of Binary 1:N Relationship Types.

Step 5: Mapping of Binary M:N Relationship Types.

Step 6: Mapping of Multivalued attributes.

Step 7: Mapping of N-ary Relationship Types.

- **Mapping EER Model Constructs to Relations**

Step 8: Options for Mapping Specialization or Generalization.

Step 9: Mapping of Union Types (Categories).

Mapping of Regular Entity Types.

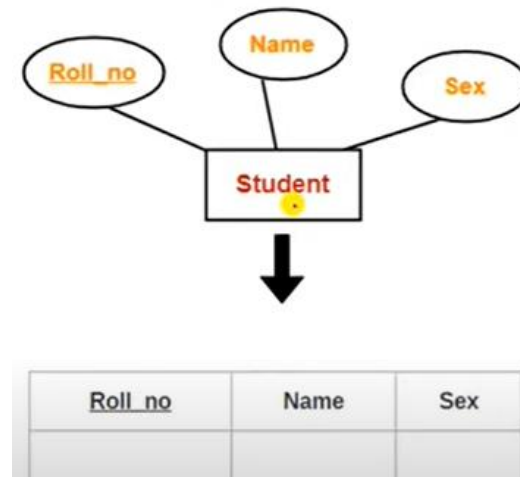
Note: FEW RANDOM EXAMPLES TO UNDERSTAND

- For each regular (strong) entity type E in the ER schema, create a relation R that includes all the simple attributes of E.
- Choose one of the key attributes of E as the primary key for R. If the chosen key of E is composite, the set of simple attributes that form it will together form the primary key of R.

: Strong Entity Set With Only Simple Attributes

A strong entity set with only *simple attributes* will require only **one table** in relational model.

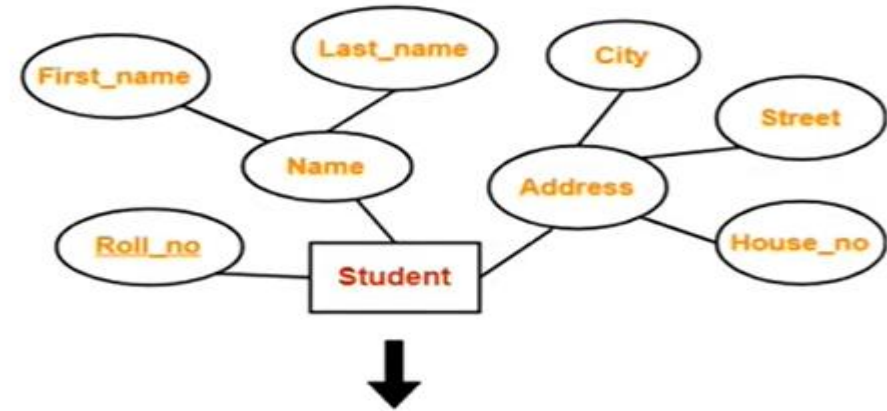
- ▣ Attributes of the table will be the attributes of the entity set.
- ▣ The primary key of the table will be the key attribute of the entity set.



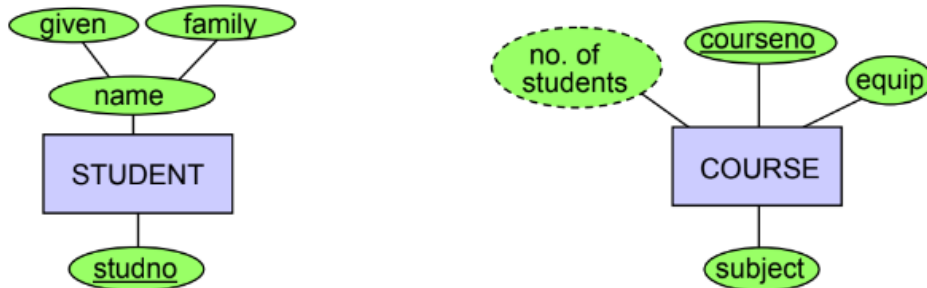
EXAMPLES:

: Strong Entity Set With Composite Attributes

- A strong entity set with any number of **composite attributes** will require **only one table** in relational model.
- While conversion, simple attributes of the composite attributes are taken into account and not the composite attribute itself.



<u>Roll_no</u>	First_name	Last_name	House_no	Street	City



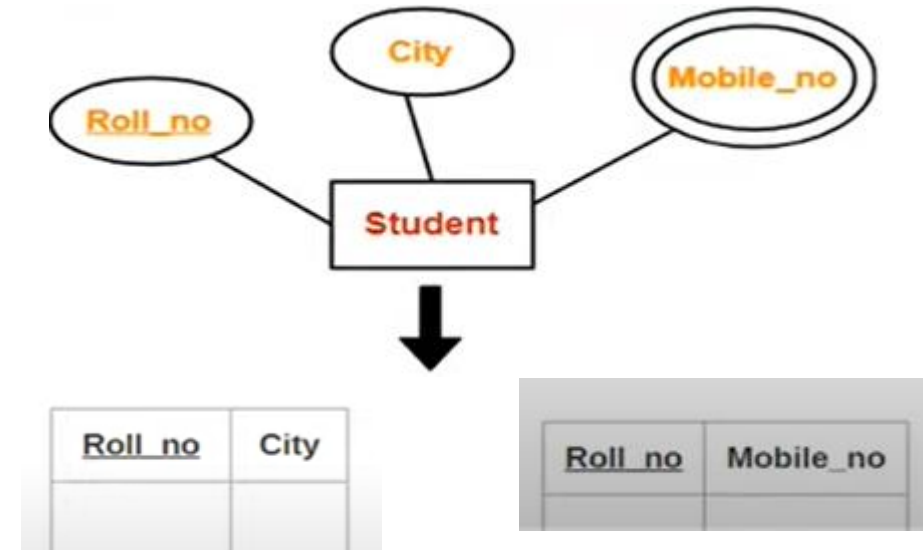
STUDENT (studno, givenname, familyname)

COURSE (courseno, subject, equip)

: Strong Entity Set With Multi Valued Attributes

A strong entity set with any number of **multi-valued attributes** will require **two tables** in relational model.

- ❑ One table will contain all the simple attributes with the primary key.
- ❑ Other table will contain the primary key and all the multi valued attributes.

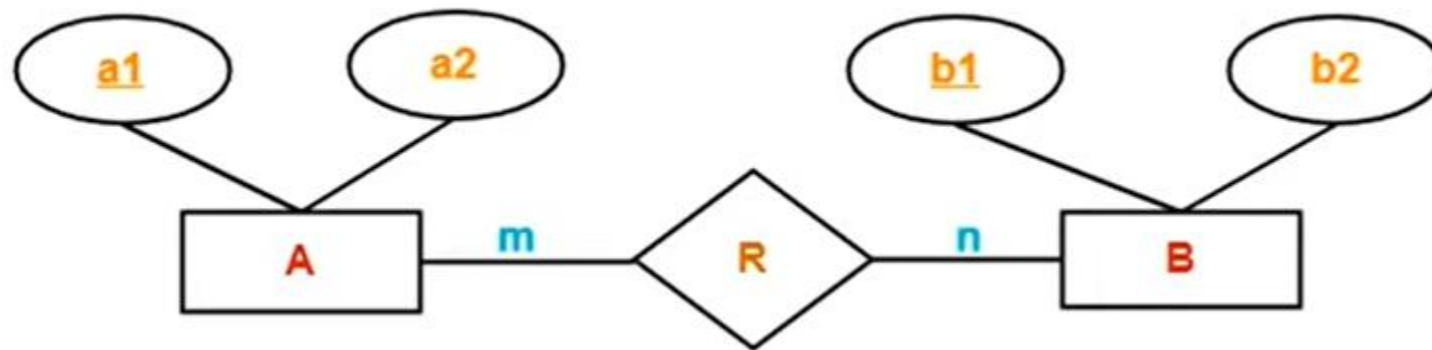


For Binary Relationships With Cardinality Ratio

- The following four cases are possible-
 - ▣ **Case-1:** Binary relationship with cardinality ratio $m:n$
 - ▣ **Case-2:** Binary relationship with cardinality ratio $1:n$
 - ▣ **Case-3:** Binary relationship with cardinality ratio $m:1$
 - ▣ **Case-4:** Binary relationship with cardinality ratio $1:1$

In *Many-to-Many relationship*, three tables will be required-

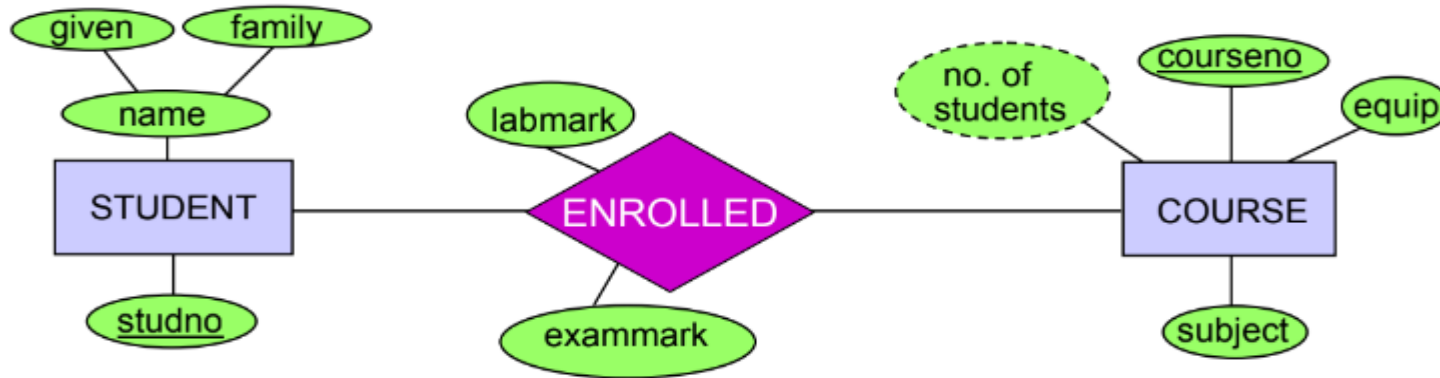
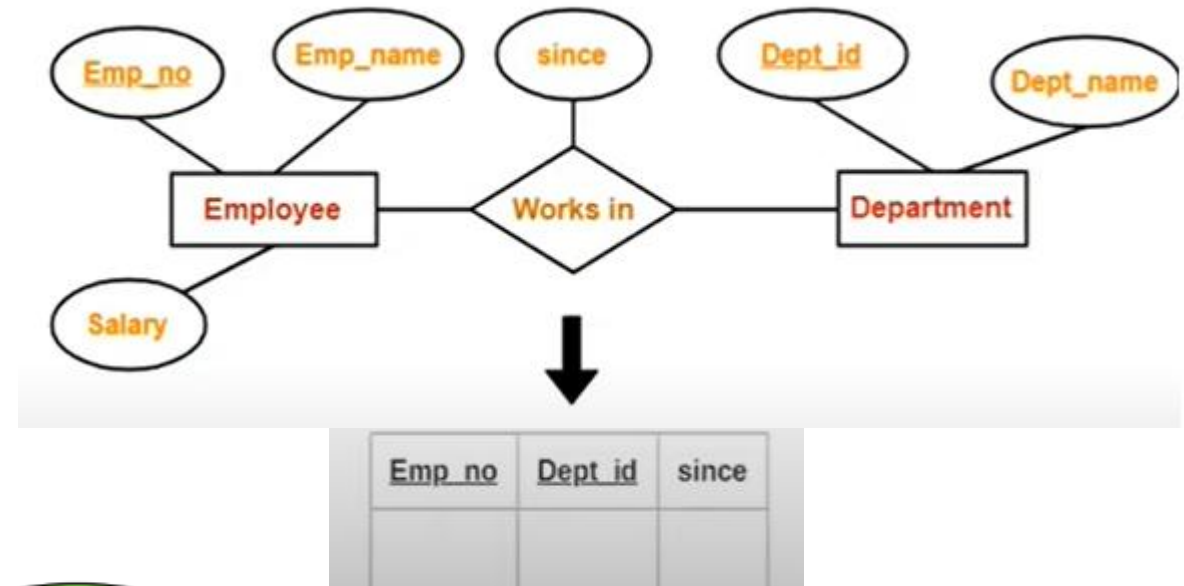
Case-1: For Binary Relationship with Cardinality Ratio $m:n$



1. $A (\underline{a1} , a2)$
2. $R (\underline{a1} , \underline{b1})$
3. $B (\underline{b1} , b2)$

: Translating Relationship Set into a Table

- A **relationship set** will require **one table** in the relational model.
- Attributes of the table are:
 - Primary key attributes of the participating entity sets
 - Its own descriptive attributes if any.
- Set of non-descriptive attributes will be the primary key

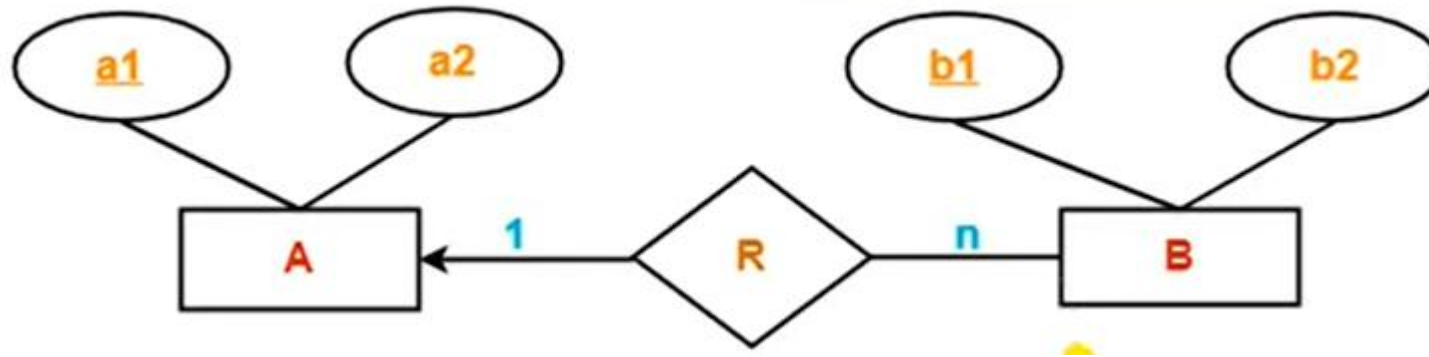


ENROL(studno, courseno, labmark, exammark)

Foreign Key ENROL(studno) references STUDENT(studno)

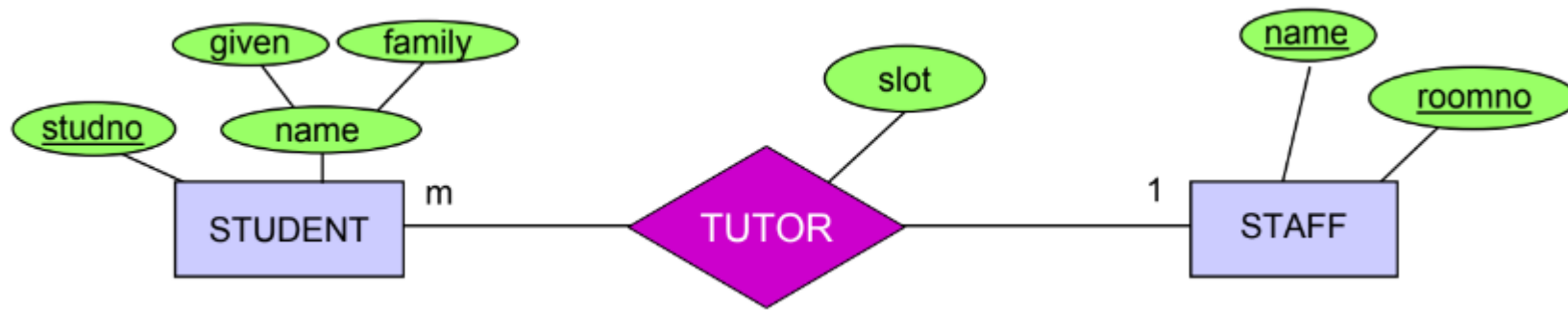
Foreign Key ENROL(courseno) references COURSE(courseno)

Case-2: For Binary Relationship with Cardinality Ratio 1:n



In **One-to-Many relationship**, two tables will be required-

1. $A (\underline{a1} , a2)$
2. $BR (\underline{b1} , b2, a1)$



STUDENT(studno, givenname, familyname)

is extended to

STUDENT(studno, givenname, familyname, **tutor**, **roomno**, **slot**)

and the constraint

Foreign Key STUDENT(tutor,roomno) references STAFF(name,roomno)

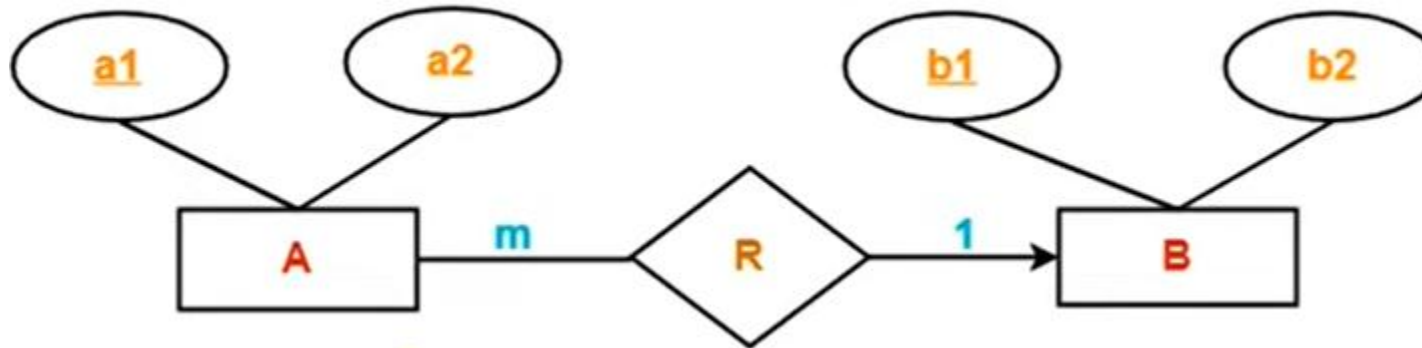
STUDENT

<u>studno</u>	given	family	tutor	roomno	slot
s1	fred	jones	bush	2.26	12B
s2	mary	brown	kahn	IT206	12B
s3	sue	smith	goble	2.82	10A
s4	fred	bloggs	goble	2.82	11A
s5	peter	jones	zobel	2.34	13B
s6	jill	peters	kahn	IT206	12A

STAFF

<u>name</u>	<u>roomno</u>
kahn	IT206
bush	2.26
goble	2.82
zobel	2.34
watson	IT212
woods	IT204
capon	A14
lindsey	2.10
barringer	2.125

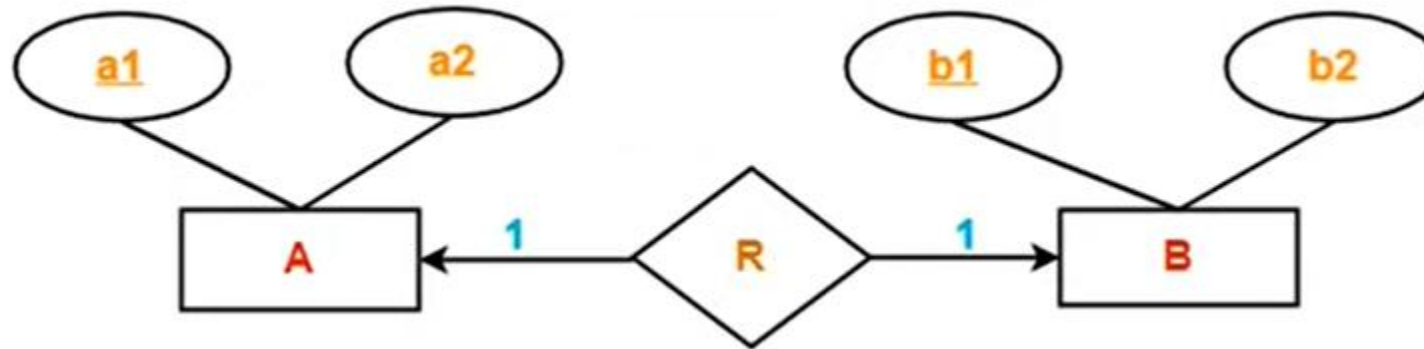
Case-3: For Binary Relationship with Cardinality Ratio m:1



In **Many-to-One relationship**, two tables will be required-

1. $AR (\underline{a1} , a2 , b1)$
2. $B (\underline{b1} , b2)$

Case-4: For Binary Relationship with Cardinality Ratio 1:1



In **One-to-One relationship**, two tables will be required. Either combine 'R' with 'A' or 'B'

AR (a1 , a2 , b1)
B (b1 , b2)

OR

A (a1 , a2)
BR (b1 , b2, a1)

: For Binary Relationship with Both Cardinality Constraints and Participation Constraints

- Cardinality constraints will be implemented as discussed in Rule-5.
- Because of the *total participation constraint*, foreign key acquires **NOT NULL** constraint i.e. now foreign key can not be null.
- Two Cases:
 - **Case-1:** For Binary Relationship with Cardinality Constraint and Total Participation Constraint from One Side
 - **Case-2:** For Binary Relationship with Cardinality Constraint and Total Participation Constraint from Both Sides

Partial participation



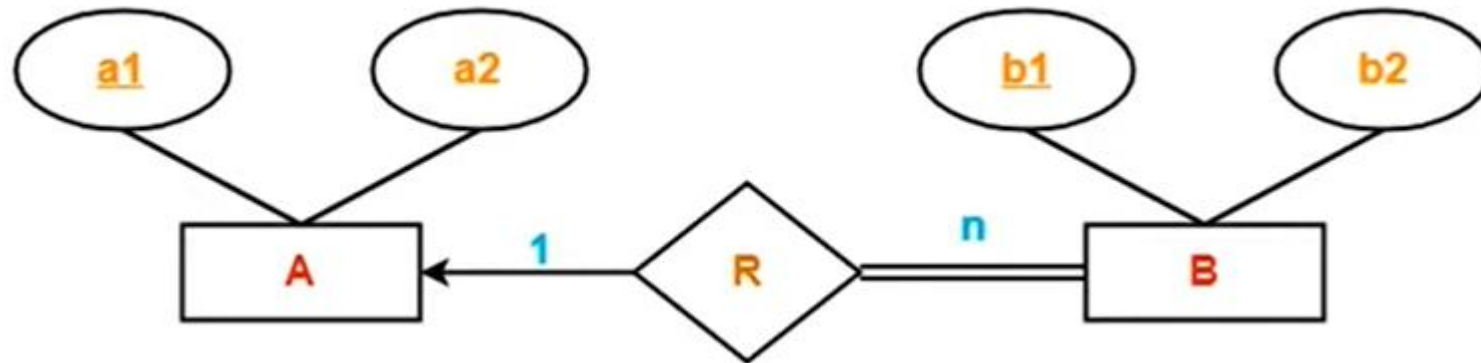
<u>EID</u>	Name	Qualification
1002	Ashwini Kunte	PhD
1003	Mita B	PhD
1007	Tanuja Sarode	PhD
1004	Madhuri Maam	PhD
1111	Ajay Vijay	ME

<u>EID</u>	<u>DID</u>
1002	40
1003	30
1007	10
1004	20

<u>DID</u>	Dname	Location
10	COMPS	NB1FLOOR
20	IT	NB8FLOOR
30	BIOMED	OB3FLOOR
40	EXTC	NB6FLOOR
50	CHEMICAL	OB5FLOOR

Note: In this case(Partial participation and 1:1 cardinality) Needs to create two tables and not three

Case-1: For Binary Relationship with Cardinality Constraint and Total Participation Constraint from One Side



Because cardinality ratio = $1 : n$, we will combine the entity set B and relationship set R .

Then, **two tables** will be required-

1. $A (\underline{a1} , a2)$
2. $BR (\underline{b1} , b2 , a1)$

Total Participation



<u>EID</u>	Name	Qualification
1002	Ashwini Kunte	PhD
1003	Mita B	PhD
1007	Tanuja Sarode	PhD
1004	Madhuri Maam	PhD
1111	Ajay Vijay	ME

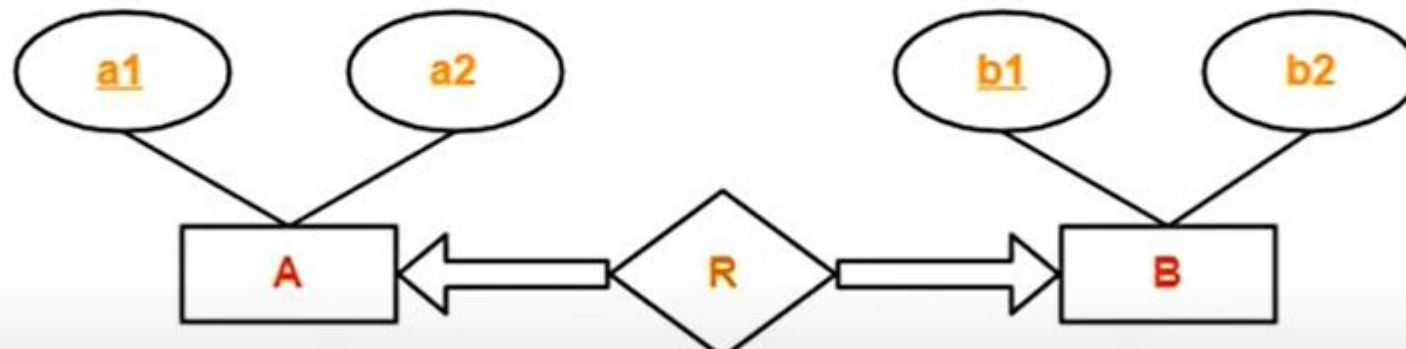
<u>EID</u>	<u>DID</u>
1002	40
1003	30
1007	10
1004	20
1111	50

<u>DID</u>	Dname	Location
10	COMPS	NB1FLOOR
20	IT	NB8FLOOR
30	BIOMED	OB3FLOOR
40	EXTC	NB6FLOOR
50	CHEMICAL	OB5FLOOR

Note: In this case(total participation and 1:1 cardinality)Needs to create One table and not three

Case-2: For Binary Relationship with Cardinality Constraint and Total Participation Constraint from Both Sides

- If there is a key constraint from both the sides of an entity set with total participation, then that binary relationship is represented using **only single table**.



Here, Only **one table** is required.

1. **ARB (a1 , a2 , b1 , b2)**

: For Binary Relationship with Weak Entity Set

- Weak entity set always appears in association with identifying relationship with **total** participation constraint and there is always **1: n** relationship from identifying entity set to weak entity set.

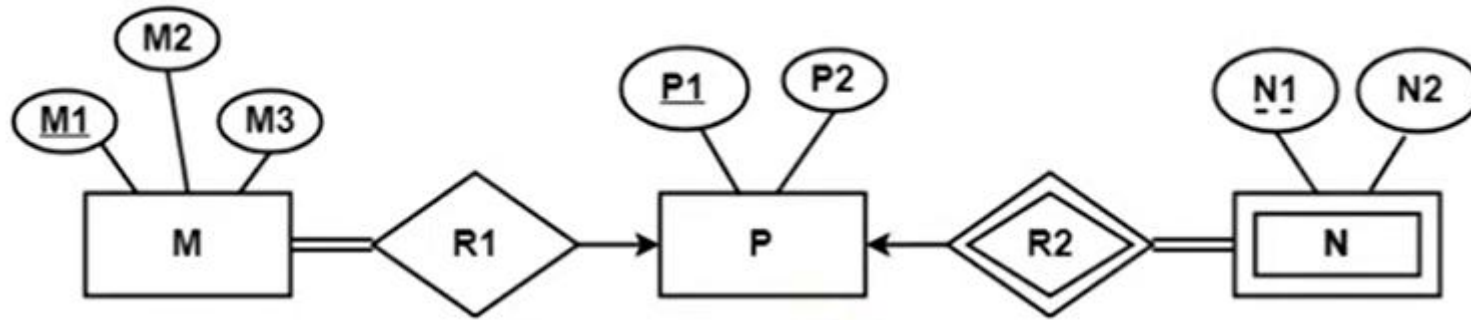


Here, **two tables** will be required-

- A (a1 , a2)**
- BR (a1 , b1 , b2)**

Example1:

- Find the minimum number of tables(relational model) required for ER diagram



- Minimum 3 tables will required

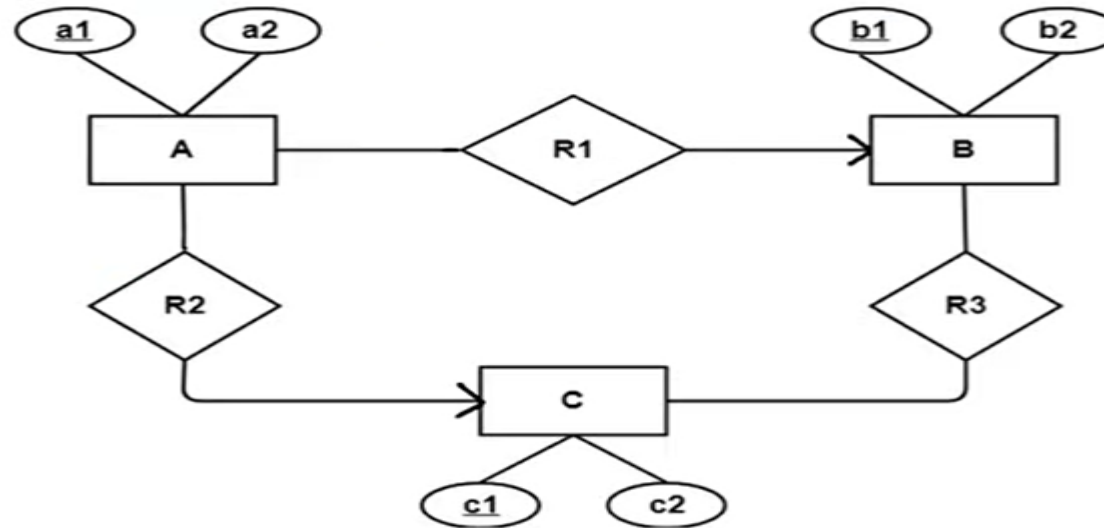
MR1 (M1 , M2 , M3 , P1)

P (P1 , P2)

NR2 (P1 , N1 , N2)

Example2:

- Find the minimum number of tables(relational model) required for ER diagram



- Minimum 4 tables will required

1. AR1R2 (a1 , a2 , b1 , c1)

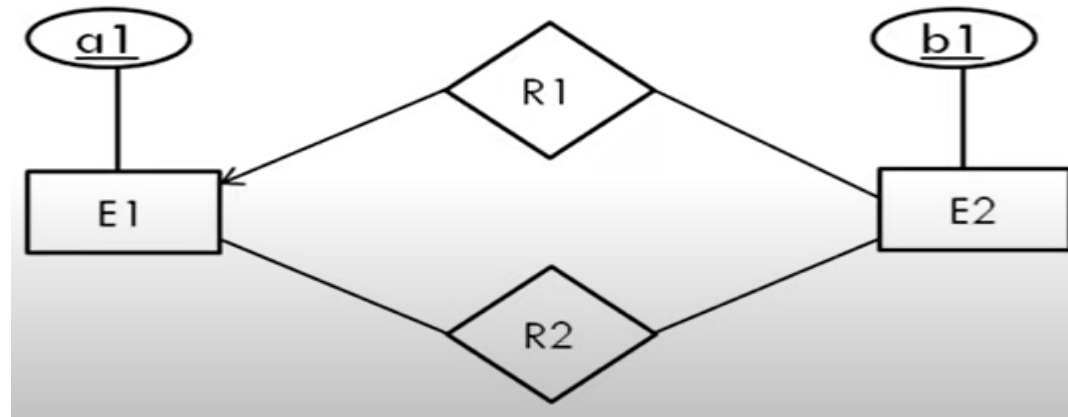
2. B (b1 , b2)

3. C (c1 , c2)

4. R3 (b1 , c1)

Example3:

- Find the minimum number of tables(relational model) required for ER diagram

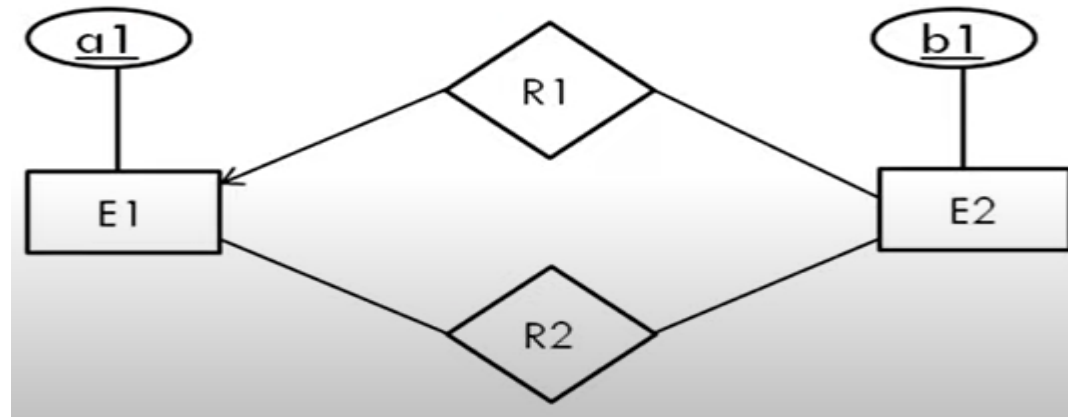


- Minimum 3 tables will required

1. E1(a1)
2. E2R1 (b1, a1)
3. R2 (a1, b1)

Example3:

- Find the minimum number of tables(relational model) required for ER diagram

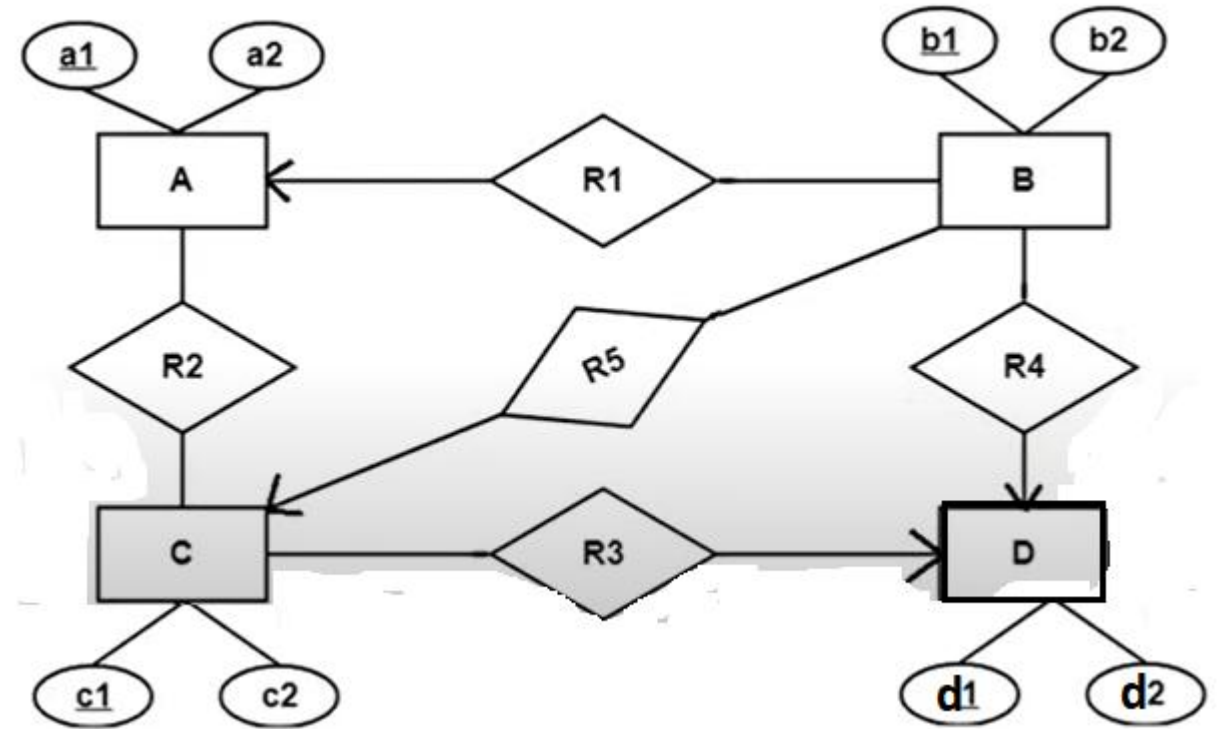


- Minimum 3 tables will required

1. `E1(a1)`
2. `E2R1 (b1, a1)`
3. `R2 (a1, b1)`

Example4:

- Find the minimum number of tables (relational model) required for ER diagram

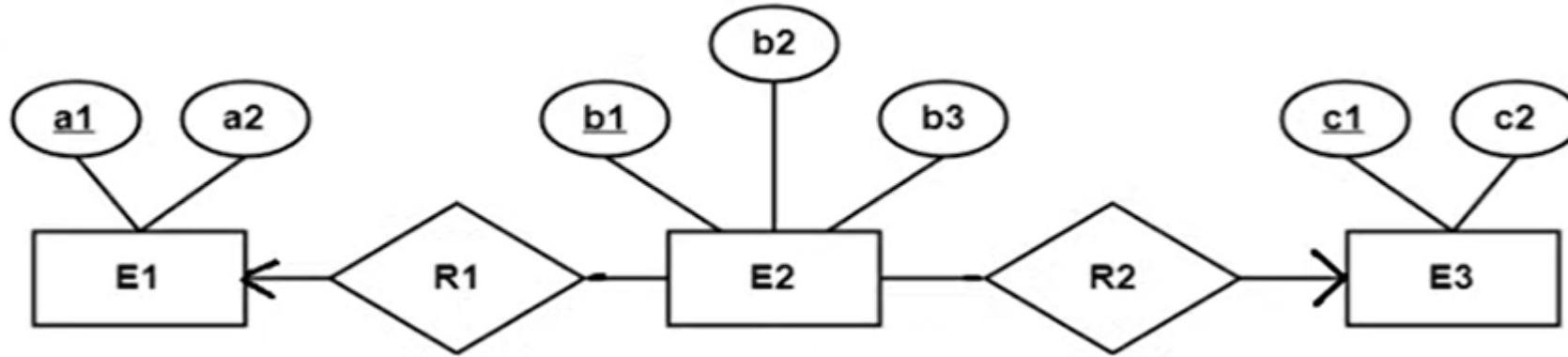


- Minimum 5 tables will required

- BR1R4R5 (b1 , b2 , a1 , c1 , d1)
- A (a1 , a2)
- R2 (a1 , c1)
- CR3 (c1 , c2 , d1)
- D (d1 , d2)

Example5:

- Find the minimum number of tables(relational model) required for ER diagram



- Minimum 3 tables will required

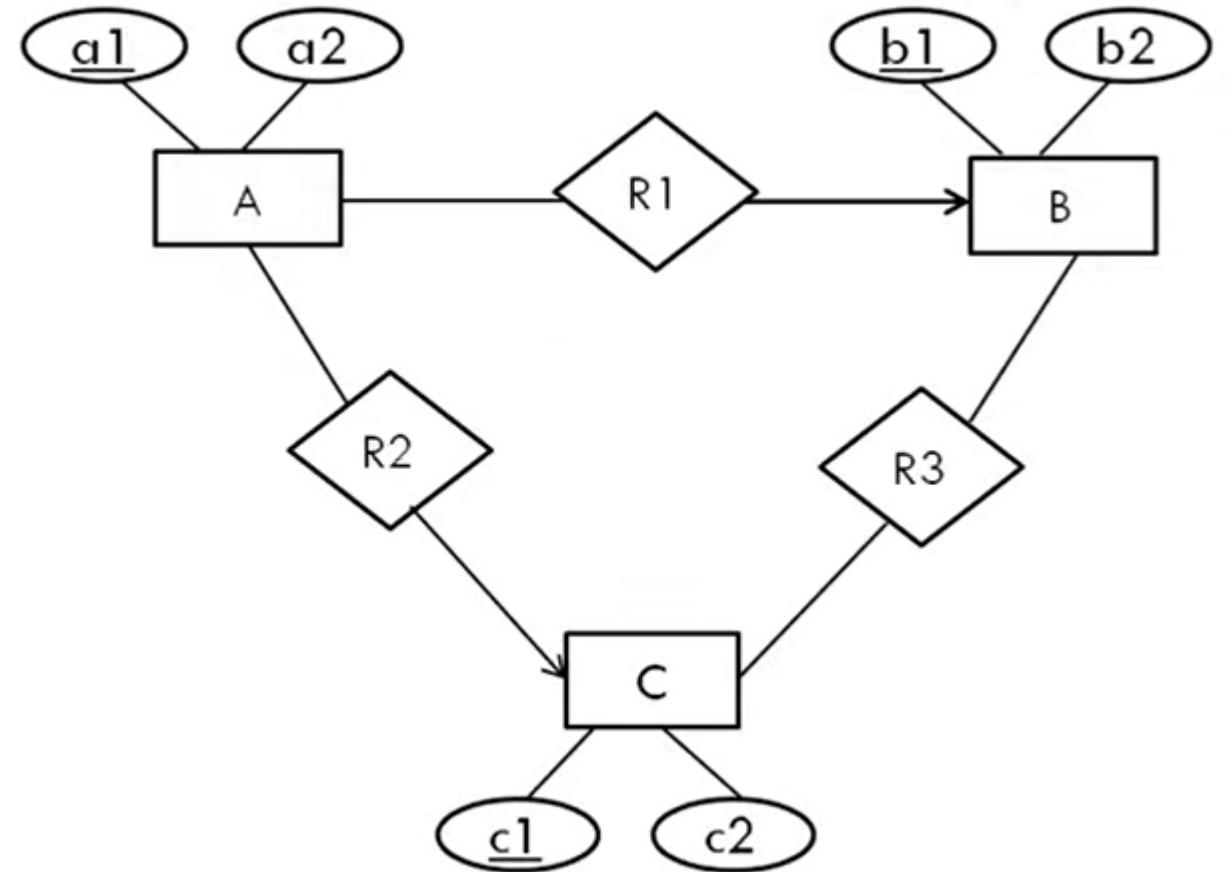
E1 (a1 , a2)

E2R1R2 (b1 , b2 , b3, a1 , c1)

E3 (c1 , c2)

Example5:

- Find the minimum number of tables(relational model) required for ER diagram



- Minimum 4 tables will required

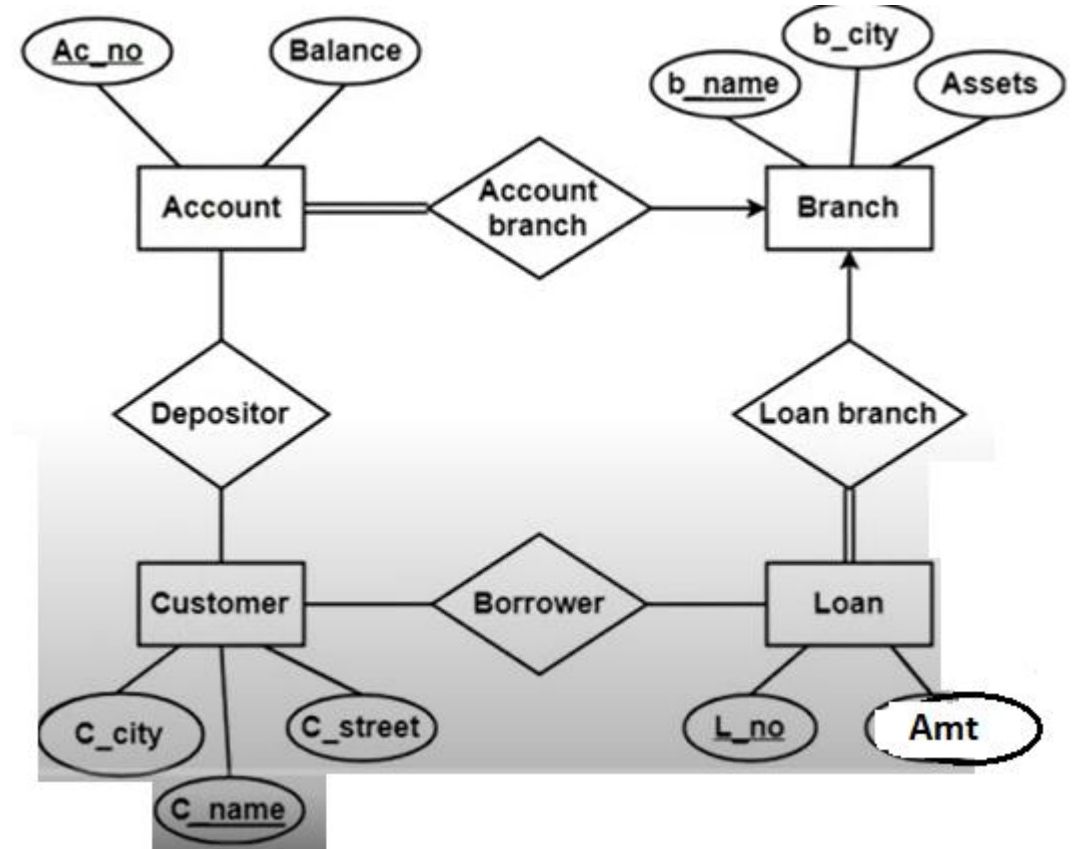
1. B (b1 , b2)
2. C (c1 , c2)
3. R3(b1, c1)
4. AR1R2 (a1 , a2, b1, c1)

Example 6:

- Find the minimum number of tables (relational model) required for ER diagram

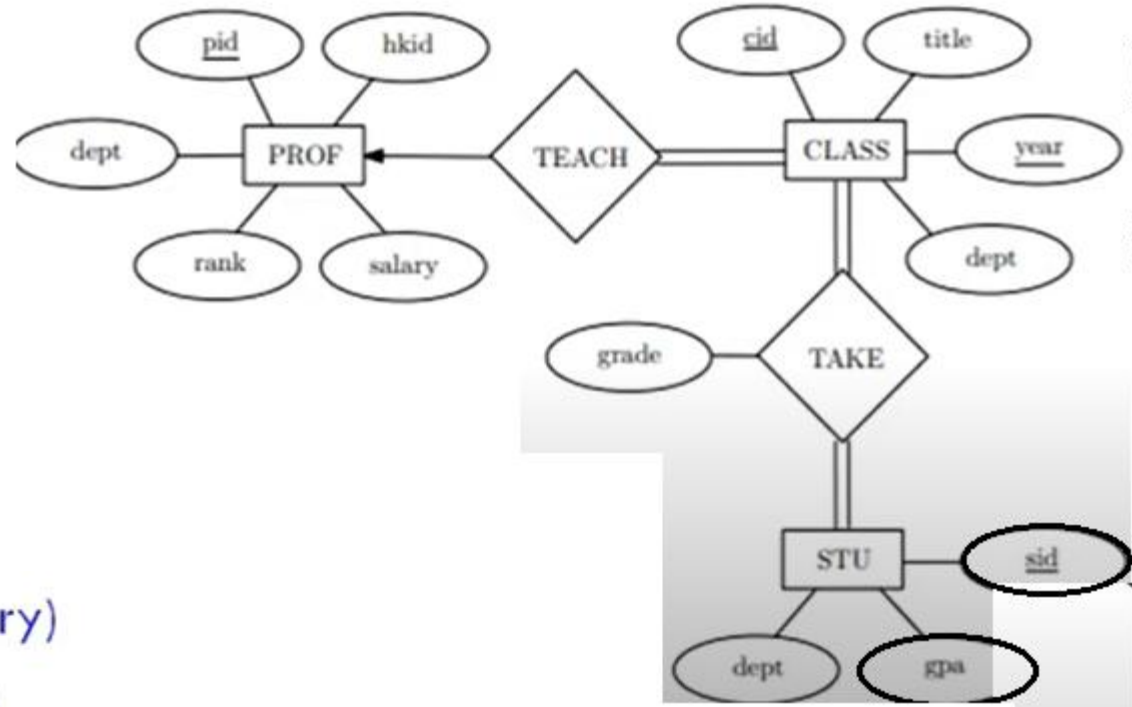
- Minimum 6 tables will be required

1. Account (Ac_no , Balance , b_name)
2. Branch (b_name , b_city , Assets)
3. Loan (L_no , Amt , b_name)
4. Borrower (C_name , L_no)
5. Customer (C_name , C_street , C_city)
6. Depositor (C_name , Ac_no)



Example 7:

- Find the minimum number of tables (relational model) required for ER diagram



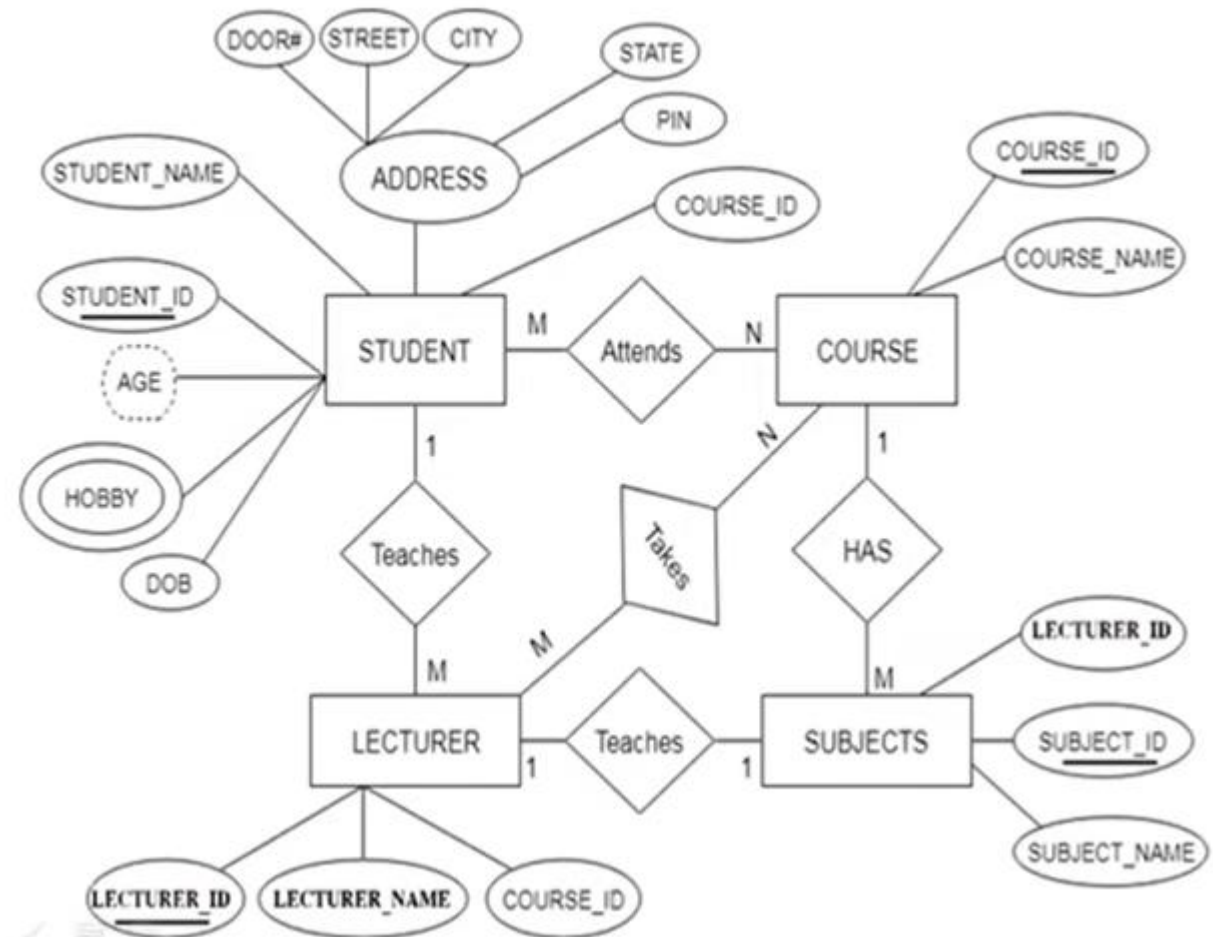
- Minimum 4 tables will require
 1. **PROF** (pid, hkid, dept, rank, salary)
 2. **CLASS** (cid, year, title, dept, pid)
 3. **STU** (sid, dept, gpa)
 4. **TAKE** (cid, year, sid, grade)

Example 8:

- Find the minimum number of tables (relational model) required for ER diagram

Minimum 7 tables will be required

1. **Student** (Student_Id, Student_Name, DOB, Door#, Street, City, State, Pin, Course_Id)
2. **Lecturer** (Lecturer_Id, Lecturer_Name, Course_Id, Student_Id, Subject_Id)
3. **Course** (Course_Id, Course_Name)
4. **Subjects** (Subject_Id, Subject_Name, Lecturer_Id, Course_Id)
5. **Hobby** (Subject_Id, Hobby)
6. **Attends** (Subject_Id, Course_Id)
7. **Takes** (Lecturer_Id, Course_Id)

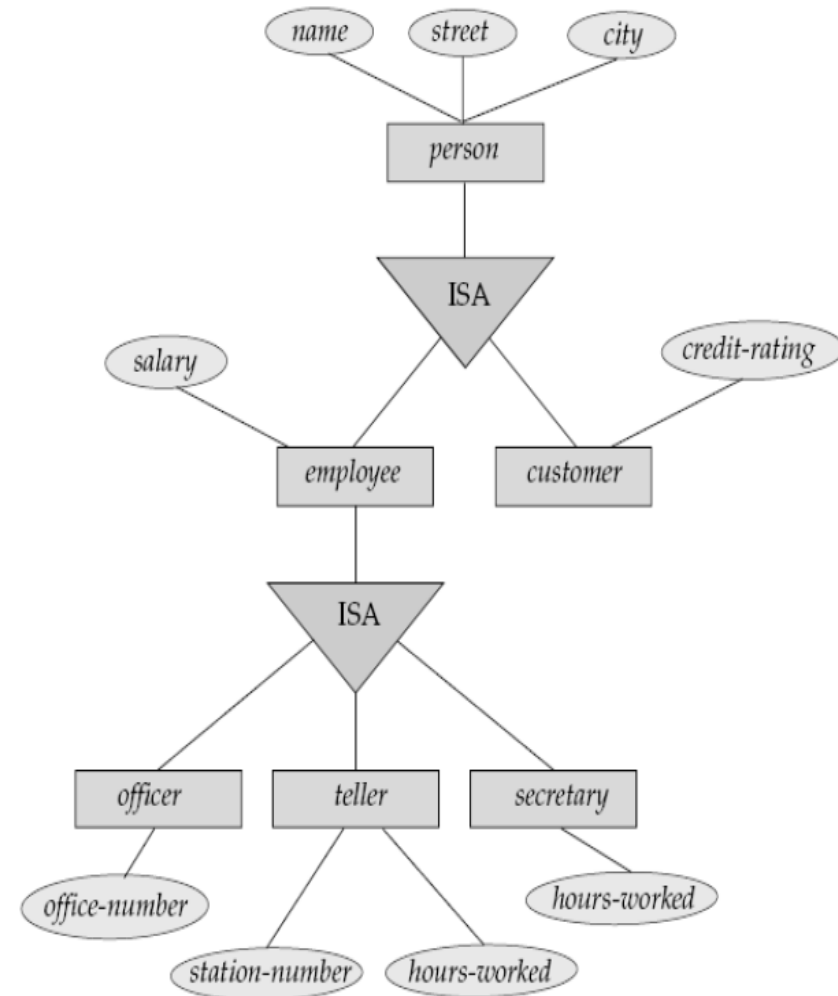


Example 9:

- Find the minimum number of tables (relational model) required for ER diagram

- Minimum 4 tables will be required

- customer** (name, street, city, credit_rating)
- officer** (name, street, city, salary, office_number)
- teller** (name, street, city, salary, station_number, hours_worked)
- secretary** (name, street, city, salary, hours_worked)



Example 10:

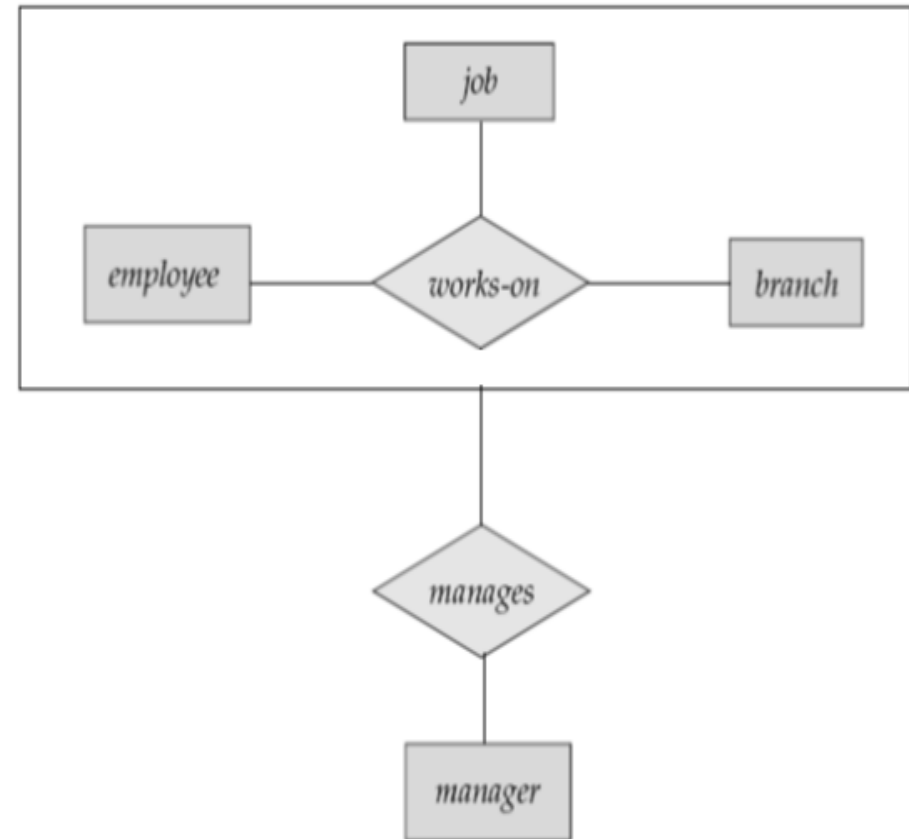
- Find the minimum number of tables(relational model) required for ER diagram
- Minimum 6 tables will required

Job schemas:

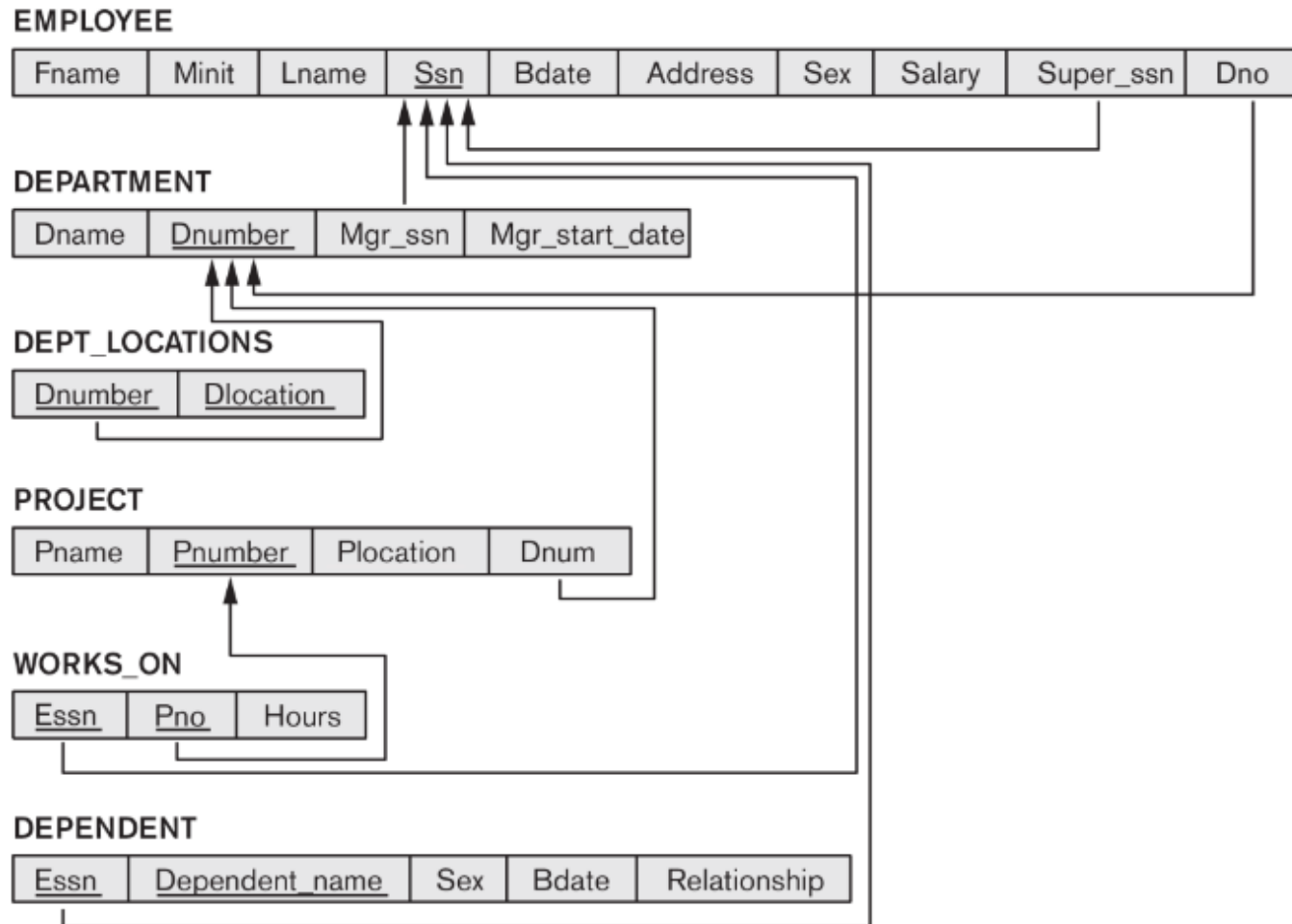
1. employee
2. job
3. Branch
4. works_on

Manager schemas:

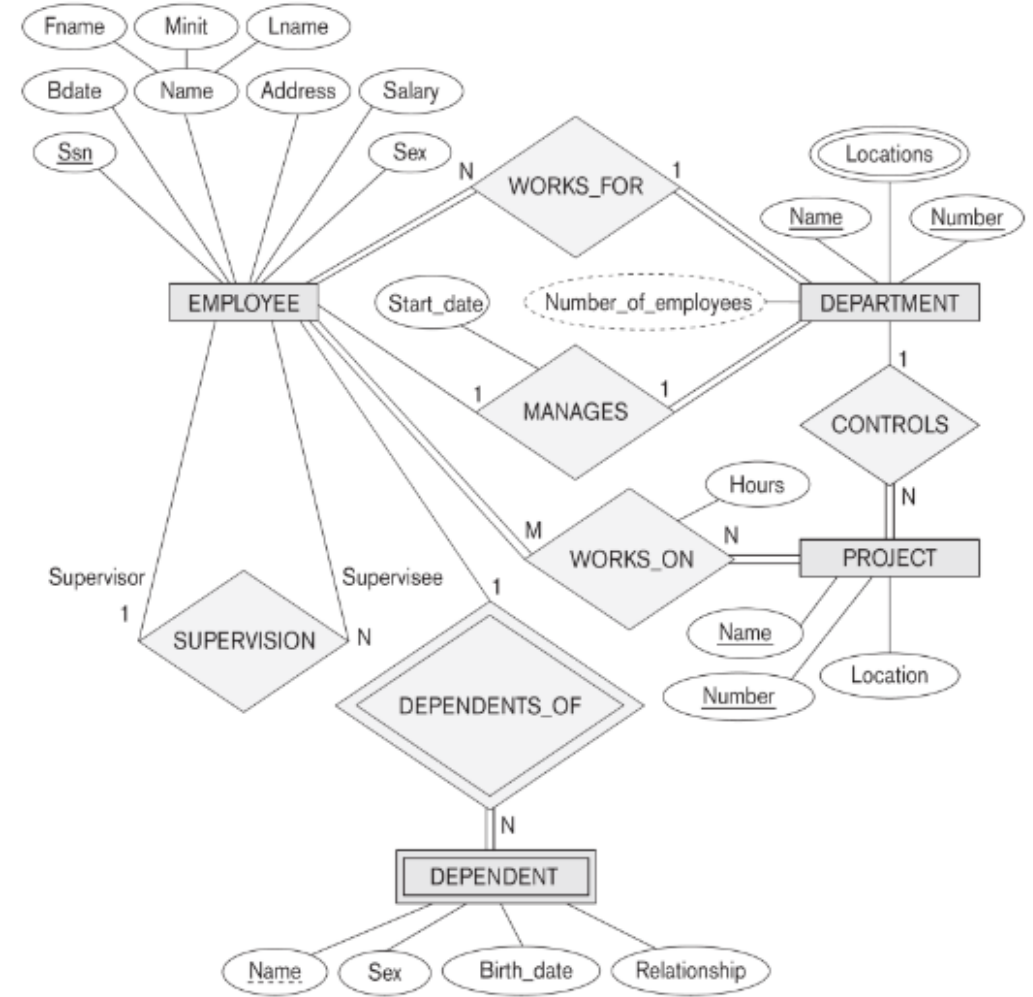
5. manager
6. manages



Example 11: Resulting Relational Schema

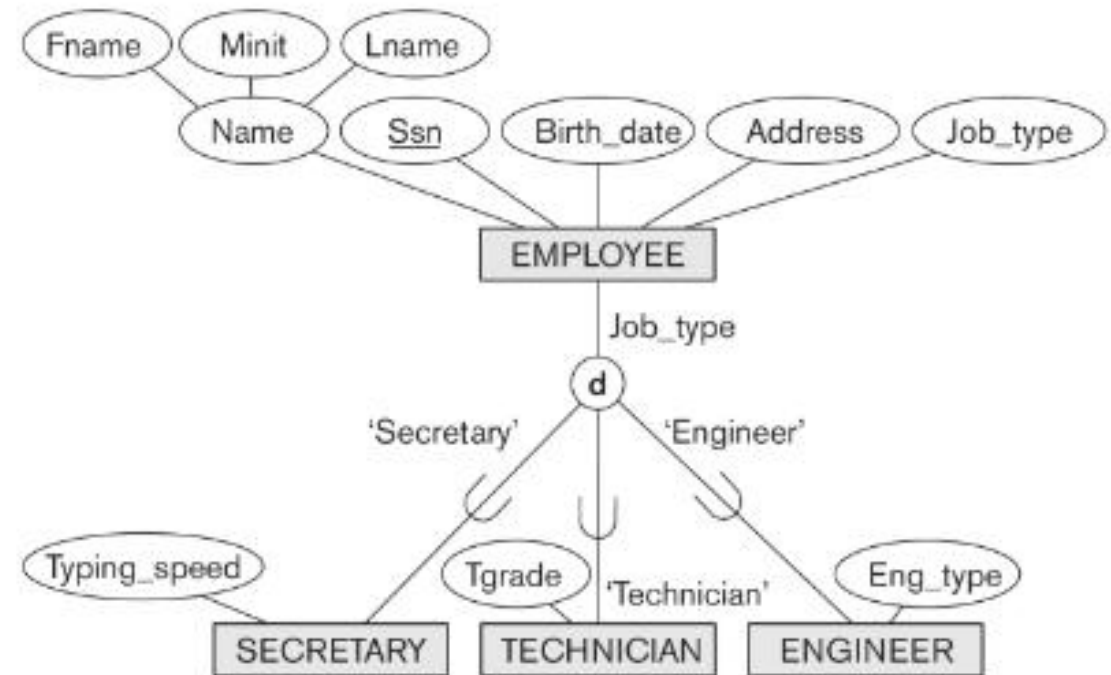
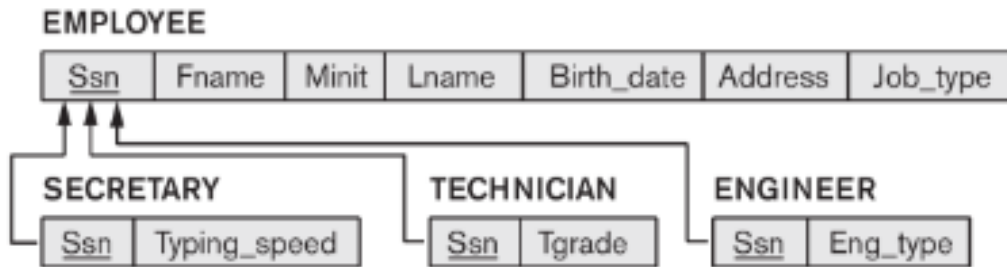


Example ERD



Example 12:

Specialization/Generalization(A)



Example 13:

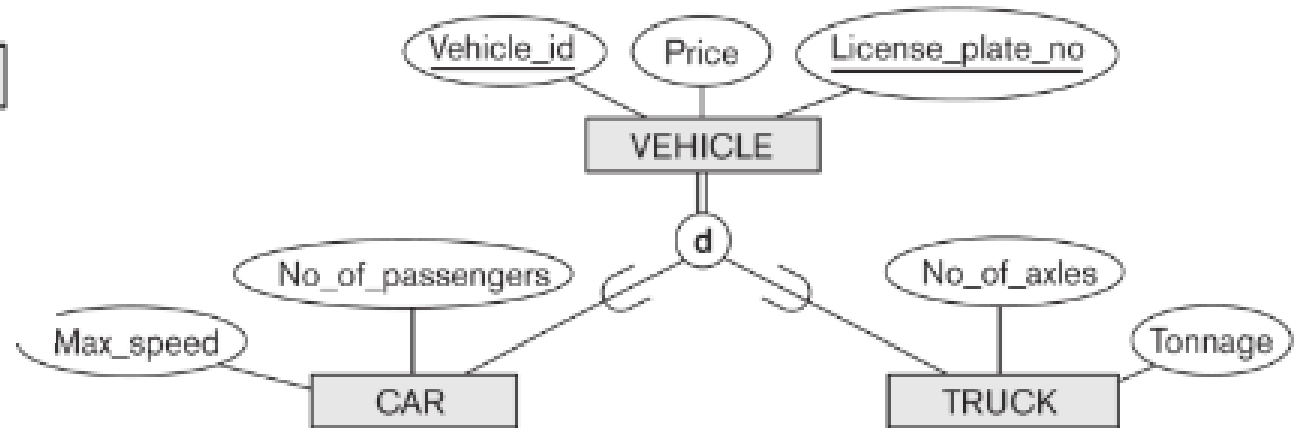
Specialization/Generalization(B)

CAR

<u>Vehicle_id</u>	License_plate_no	Price	Max_speed	No_of_passengers
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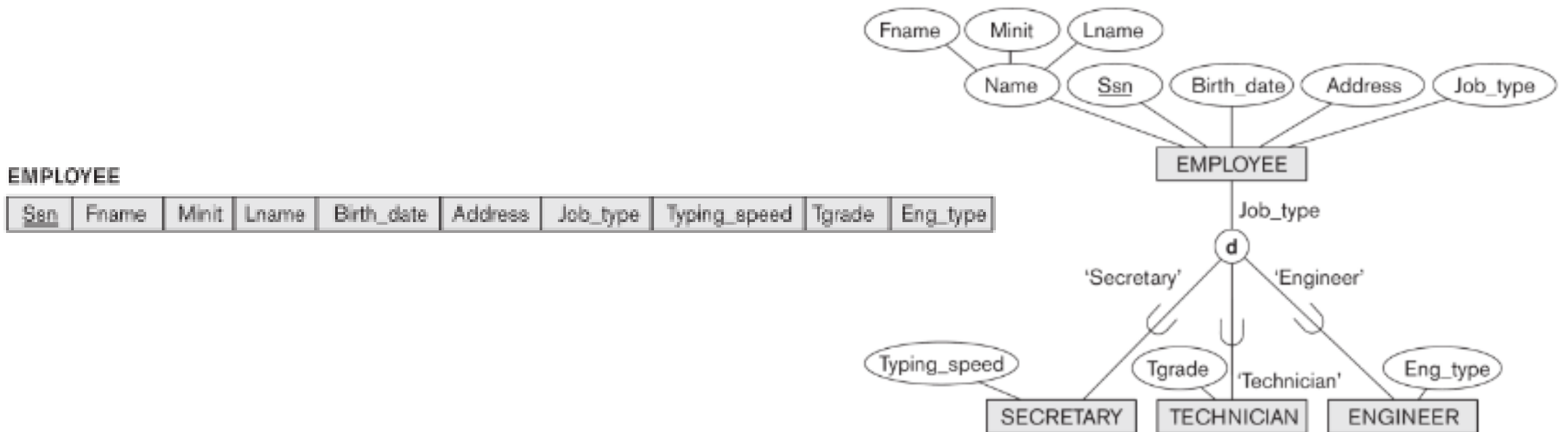
TRUCK

<u>Vehicle_id</u>	License_plate_no	Price	No_of_axles	Tonnage
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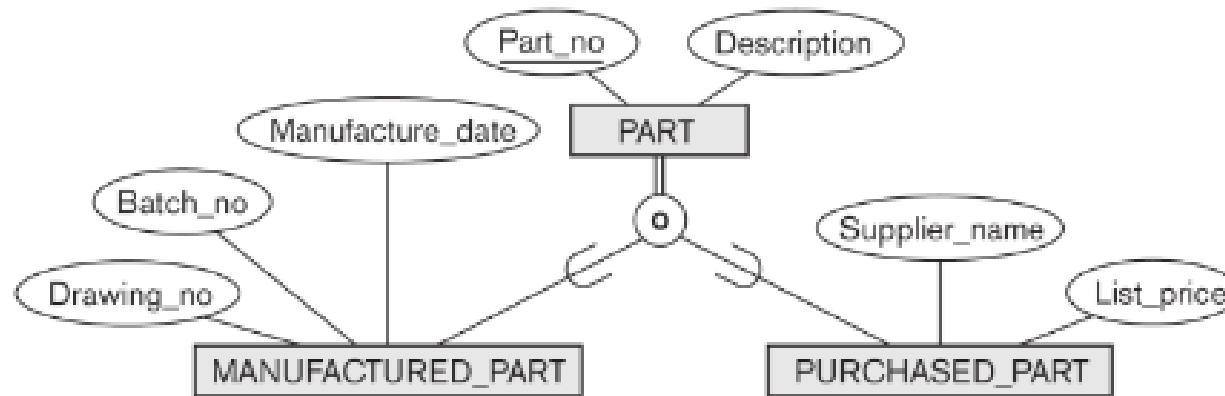
Example 14:

Specialization/Generalization(C)



Example 15:

Specialization/Generalization(D)



PART

<u>Part_no</u>	Description	Mflag	Drawing_no	Manufacture_date	Batch_no	Pflag	Supplier_name	List_price
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Steps for mapping

- **ER-to-Relational Mapping Algorithm**

Step 1: Mapping of Regular Entity Types

Step 2: Mapping of Weak Entity Types

Step 3: Mapping of Binary 1:1 Relation Types

Step 4: Mapping of Binary 1:N Relationship Types.

Step 5: Mapping of Binary M:N Relationship Types.

Step 6: Mapping of Multivalued attributes.

Step 7: Mapping of N-ary Relationship Types.

- **Mapping EER Model Constructs to Relations**

Step 8: Options for Mapping Specialization or Generalization.

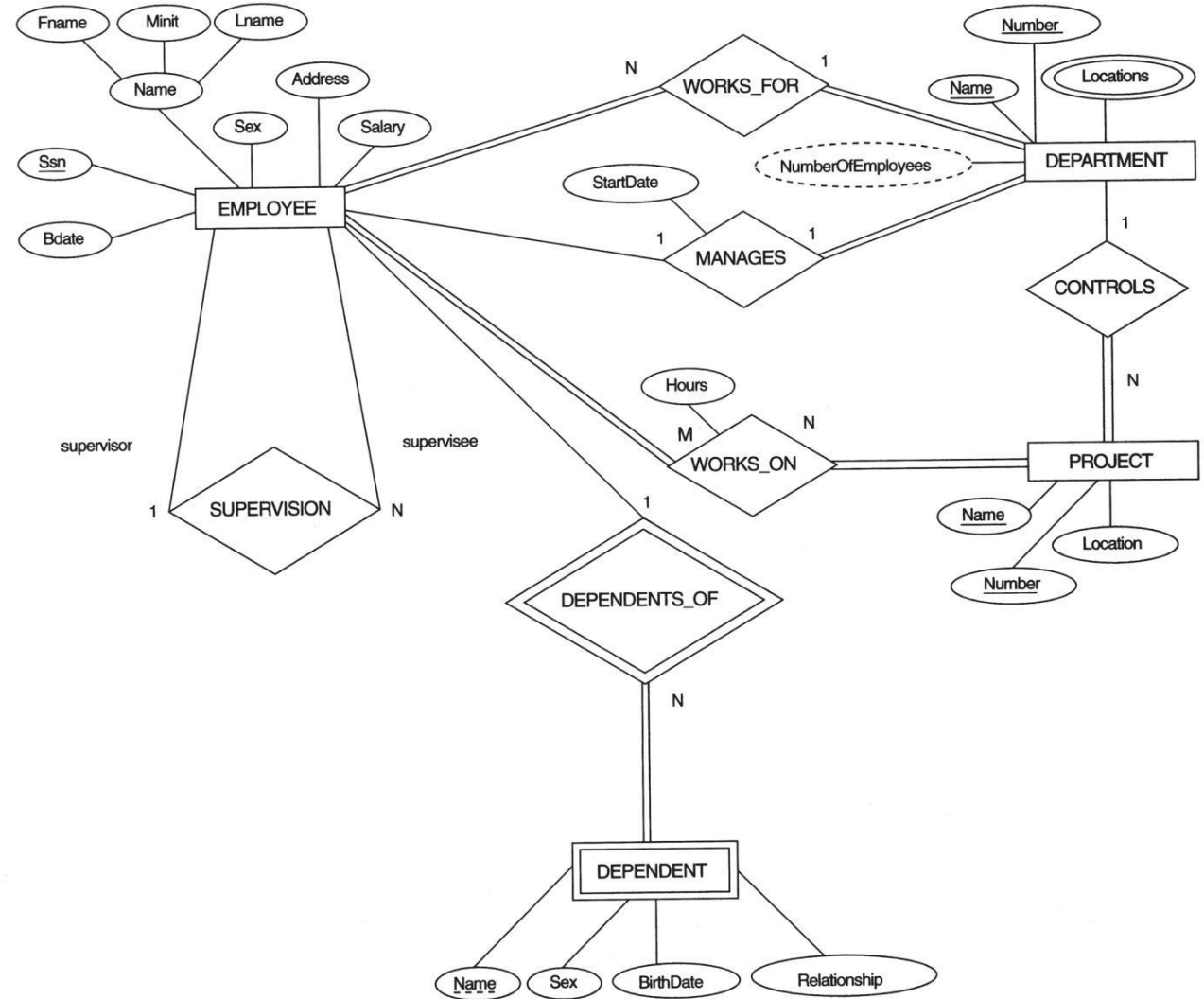
20 Step 9: Mapping of Union Types (Categories).

Step 1: Mapping of Regular Entity Types.

- For each regular (strong) entity type E in the ER schema, create a relation R that includes all the simple attributes of E .
- Choose one of the key attributes of E as the primary key for R . If the chosen key of E is composite, the set of simple attributes that form it will together form the primary key of R .

Step 1

Example: We create the relations EMPLOYEE, DEPARTMENT, and PROJECT in the relational schema corresponding to the regular entities in the ER diagram. SSN, DNUMBER, and PNUMBER are the primary keys for the relations EMPLOYEE, DEPARTMENT, and PROJECT as shown.



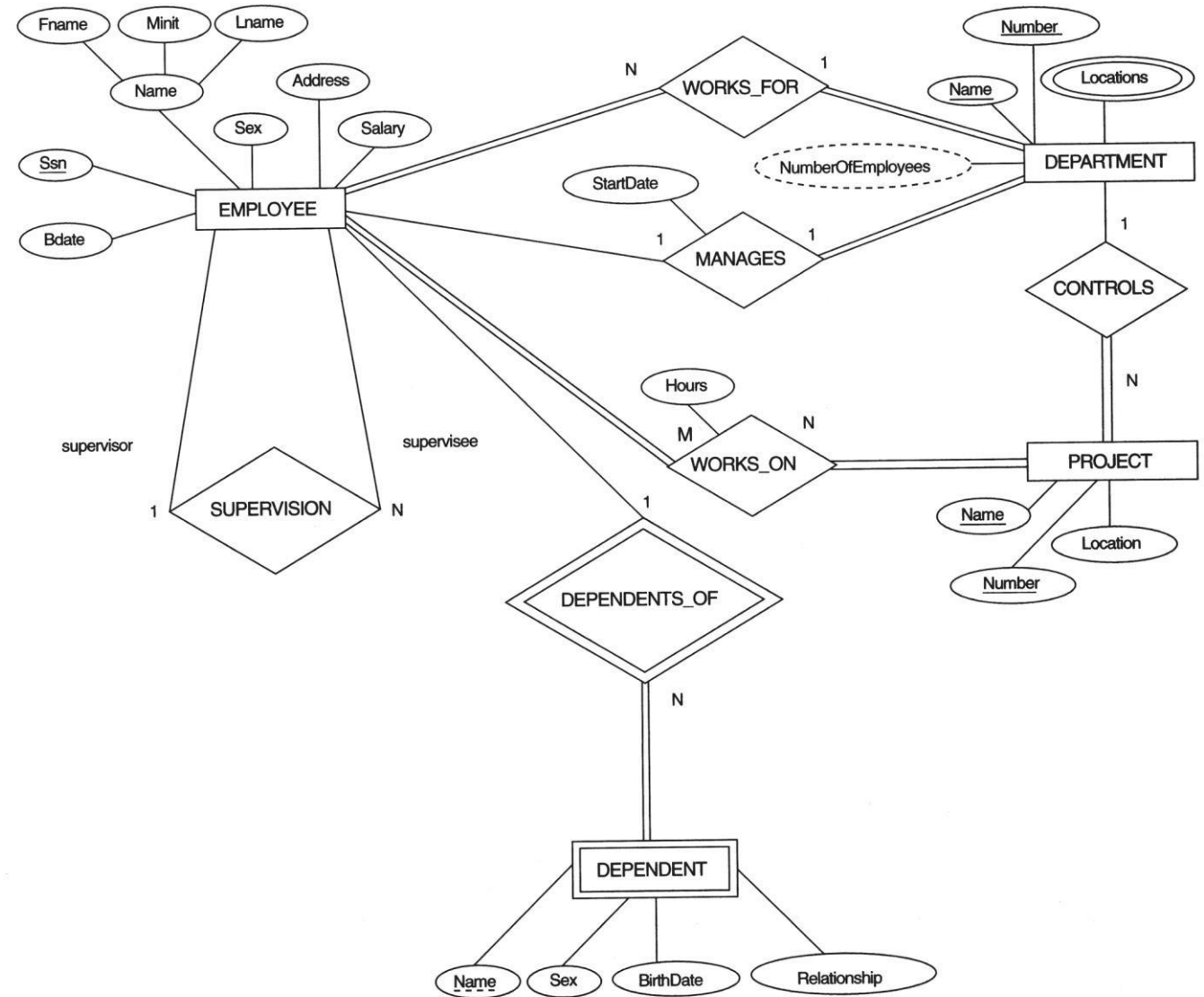
Step 2: Mapping of Weak Entity Types

- For each weak entity type W in the ER schema with owner entity type E , create a relation R and include all simple attributes (or simple components of composite attributes) of W as attributes of R .
- In addition, include as foreign key attributes of R the primary key attribute(s) of the relation(s) that correspond to the owner entity type(s).
- The primary key of R is the *combination of* the primary key(s) of the owner(s) and the partial key of the weak entity type W , if any.

Step 2

Example: Create the relation **DEPENDENT** in this step to correspond to the weak entity type **DEPENDENT**. Include the primary key **SSN** of the **EMPLOYEE** relation as a foreign key attribute of **DEPENDENT** (renamed to **ESSN**).

The primary key of the **DEPENDENT** relation is the combination {**ESSN**, **DEPENDENT_NAME**} because **DEPENDENT_NAME** is the partial key of **DEPENDENT**.



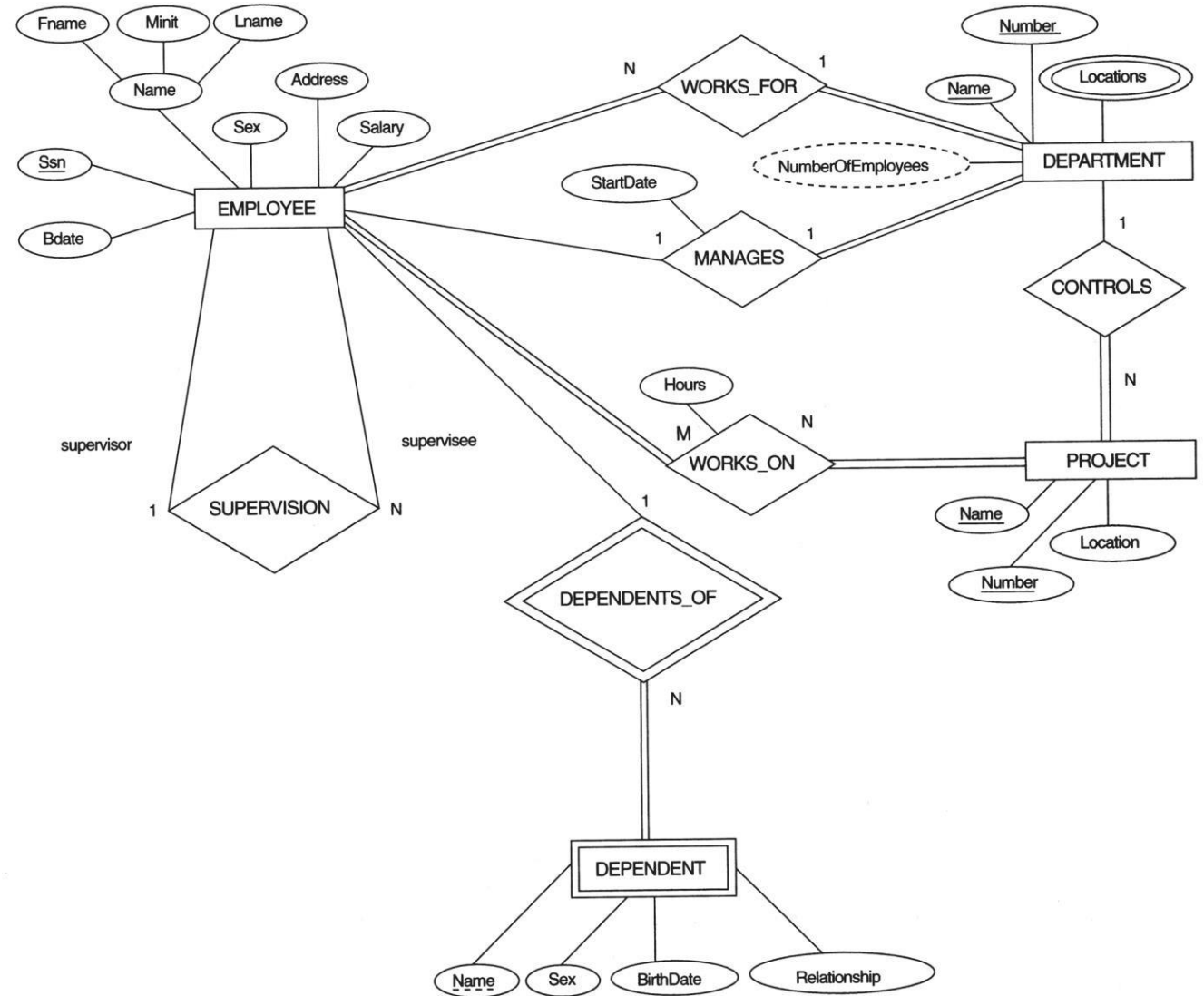
Step 3: Mapping of Binary 1:1 Relation Types

For each binary 1:1 relationship type R in the ER schema, identify the relations S and T that correspond to the entity types participating in R. There are three possible approaches:

- (1) Foreign Key approach: Choose the relations with ***total participation*** in R – say S-- and include T's primary key in S.
- (2) Merged relation option: An alternate mapping of a 1:1 relationship type is possible by merging the two entity types and the relationship into a single relation. This may be appropriate when ***both participations are total***.
- (3) Cross-reference or relationship relation option: The third alternative is to set up a third relation ***W(T.primarykey, S.primaryKey)*** for the purpose of cross-referencing the primary keys of the two relations S and T representing the entity types.

Step 3 (Foreign key approach)

Example: 1:1 relation MANAGES is mapped by choosing the participating entity type DEPARTMENT to serve in the role of S, because its participation in the MANAGES relationship type is total.

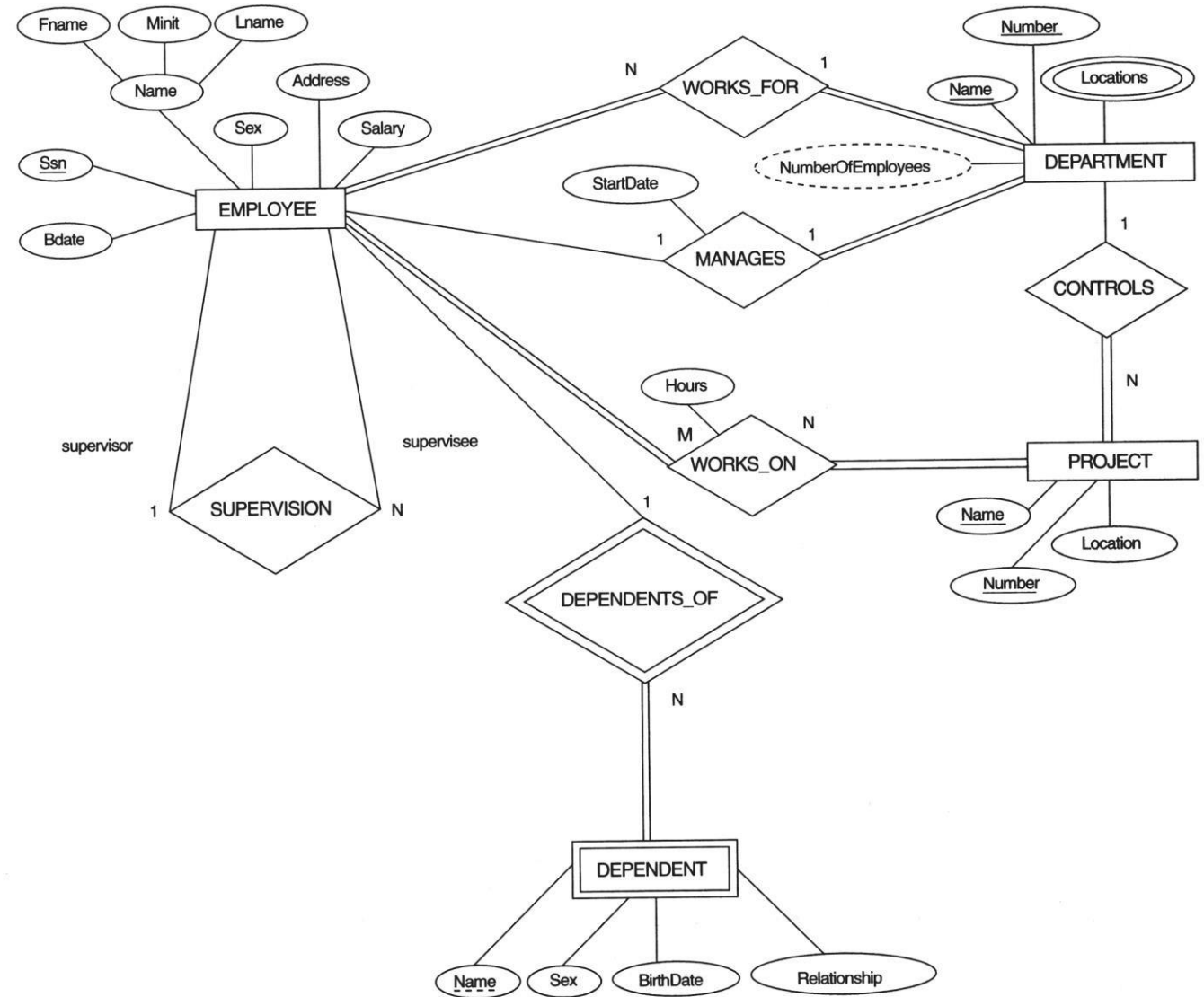


Step 4: Mapping of Binary 1:N Relationship Types

- For each regular binary 1:N relationship type R, identify the relation S that represent the participating entity type at the N-side of the relationship type.
- Include as foreign key in S the primary key of the relation T that represents the other entity type participating in R.
- Include any simple attributes of the 1:N relation type as attributes of S.

Step 4

Example: 1:N relationship types WORKS_FOR, CONTROLS, and SUPERVISION in the figure. For WORKS_FOR we include the primary key DNUMBER of the DEPARTMENT relation as foreign key in the EMPLOYEE relation and call it DNO.



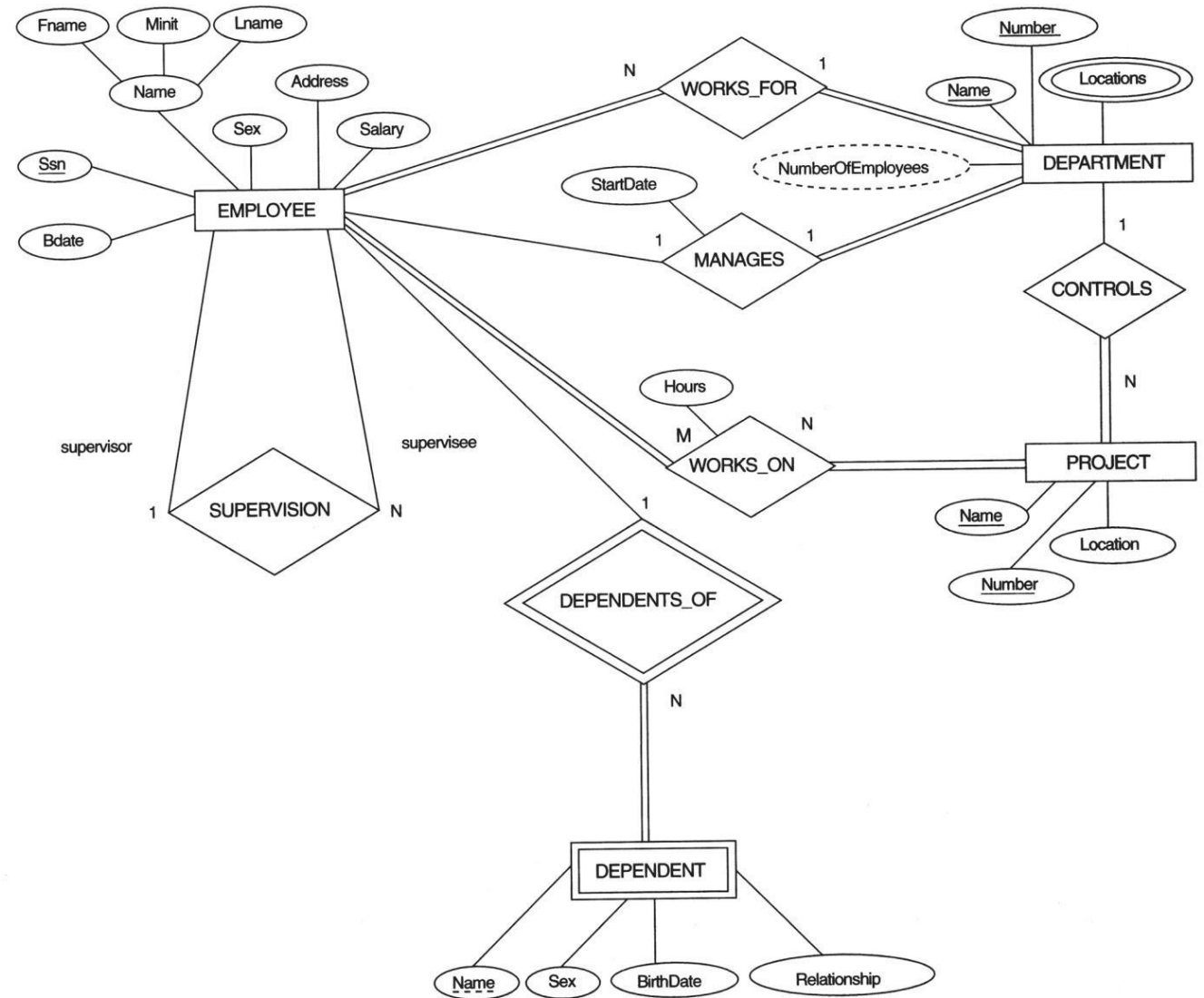
Step 5: Mapping of Binary M:N Relationship Types

- For each regular binary M:N relationship type R, *create a new relation S* to represent R.
- Include as foreign key attributes in S the primary keys of the relations that represent the participating entity types; *their combination will form the primary key* of S.
- Also include any simple attributes of the M:N relationship type (or simple components of composite attributes) as attributes of S

Step 5

Example: The M:N relationship type WORKS_ON from the ER diagram is mapped by creating a relation WORKS_ON in the relational database schema. The primary keys of the PROJECT and EMPLOYEE relations are included as foreign keys in WORKS_ON and renamed PNO and ESSN, respectively.

Attribute HOURS in WORKS_ON represents the HOURS attribute of the relation type. The primary key of the WORKS_ON relation is the combination of the foreign key attributes {ESSN, PNO}.

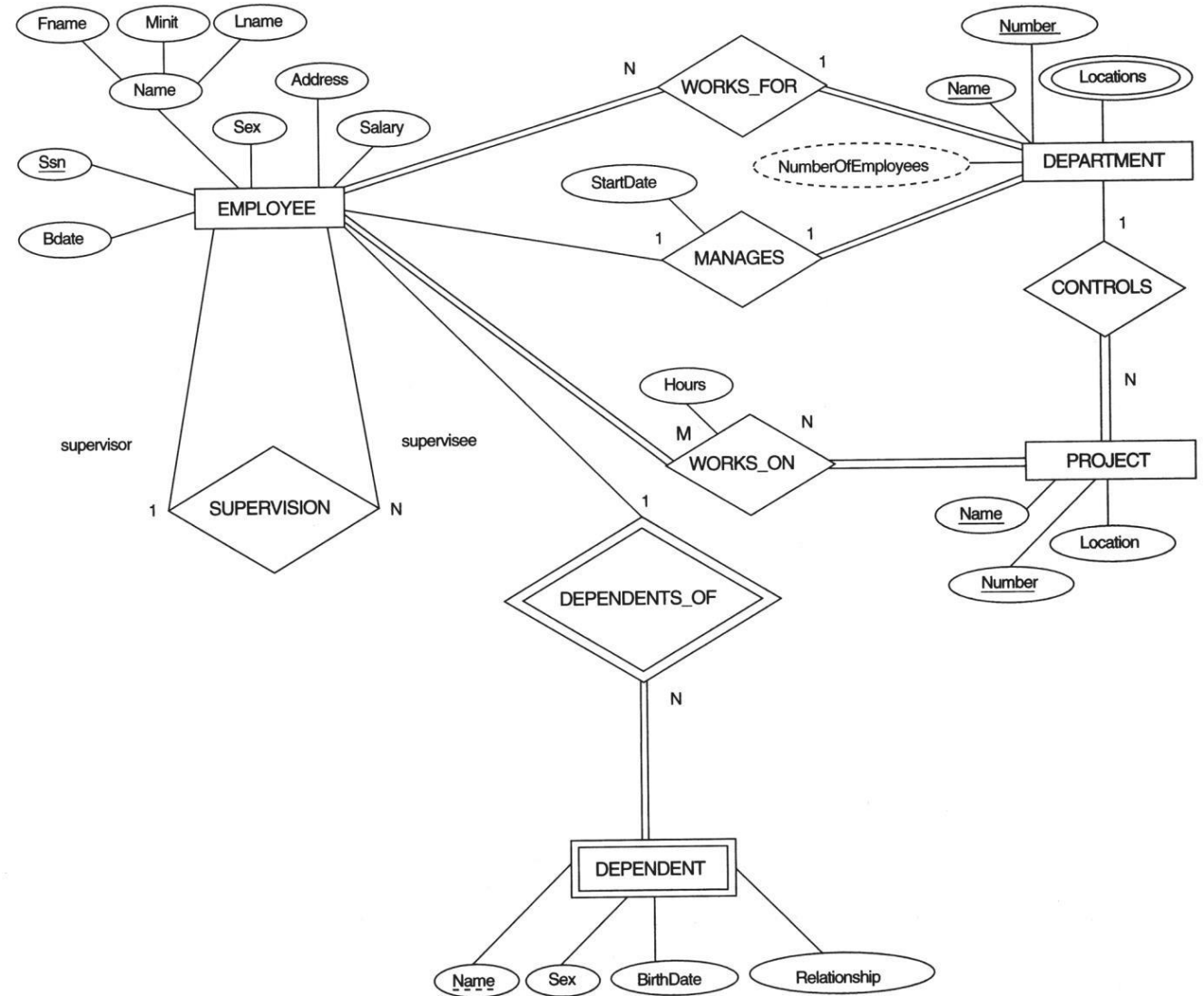


Step 6: Mapping of Multivalued attributes

- For each multivalued attribute A, create a new relation R. This relation R will include an attribute corresponding to A, plus the primary key attribute K-as a foreign key in R-of the relation that represents the entity type that has A as an attribute.
- The primary key of R is the combination of A and K. If the multivalued attribute is composite, we include its simple components.

Step 6

Example: The relation DEPT_LOCATIONS is created. The attribute DLOCATION represents the multivalued attribute LOCATIONS of DEPARTMENT, while DNUMBER-as foreign key-represents the primary key of the DEPARTMENT relation. The primary key of R is the combination of {DNUMBER, DLOCATION}.



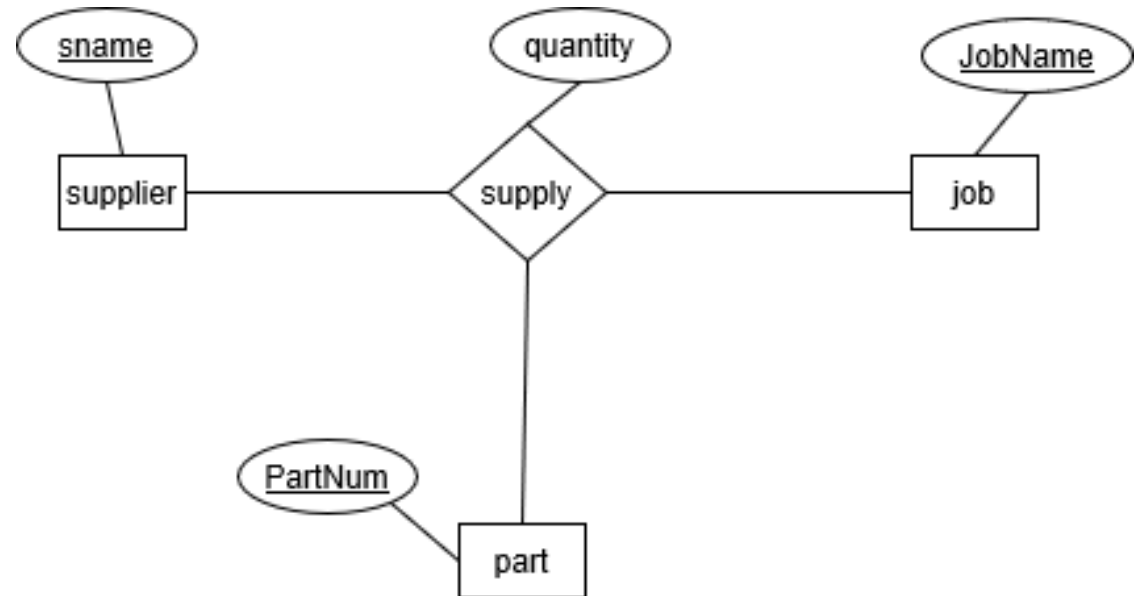
Step 7: Mapping of N-ary Relationship Types

- For each n-ary relationship type R, where $n > 2$, create a new relationship S to represent R.
- Include as foreign key attributes in S the primary keys of the relations that represent the participating entity types.
- Also include any simple attributes of the n-ary relationship type (or simple components of composite attributes) as attributes of S.

Step 7

Example: The relationship type SUPPLY in the ER

This can be mapped to the relation SUPPLY shown in the relational schema, whose primary key is the combination of the three foreign keys {SNAME, PARTNUM, JOBNAME}



Solution to n-ary example

Tables that already exist:

Supplier (sname, ...)

Job (jname, ...)

Part (partNum, ...)

Add new Table:

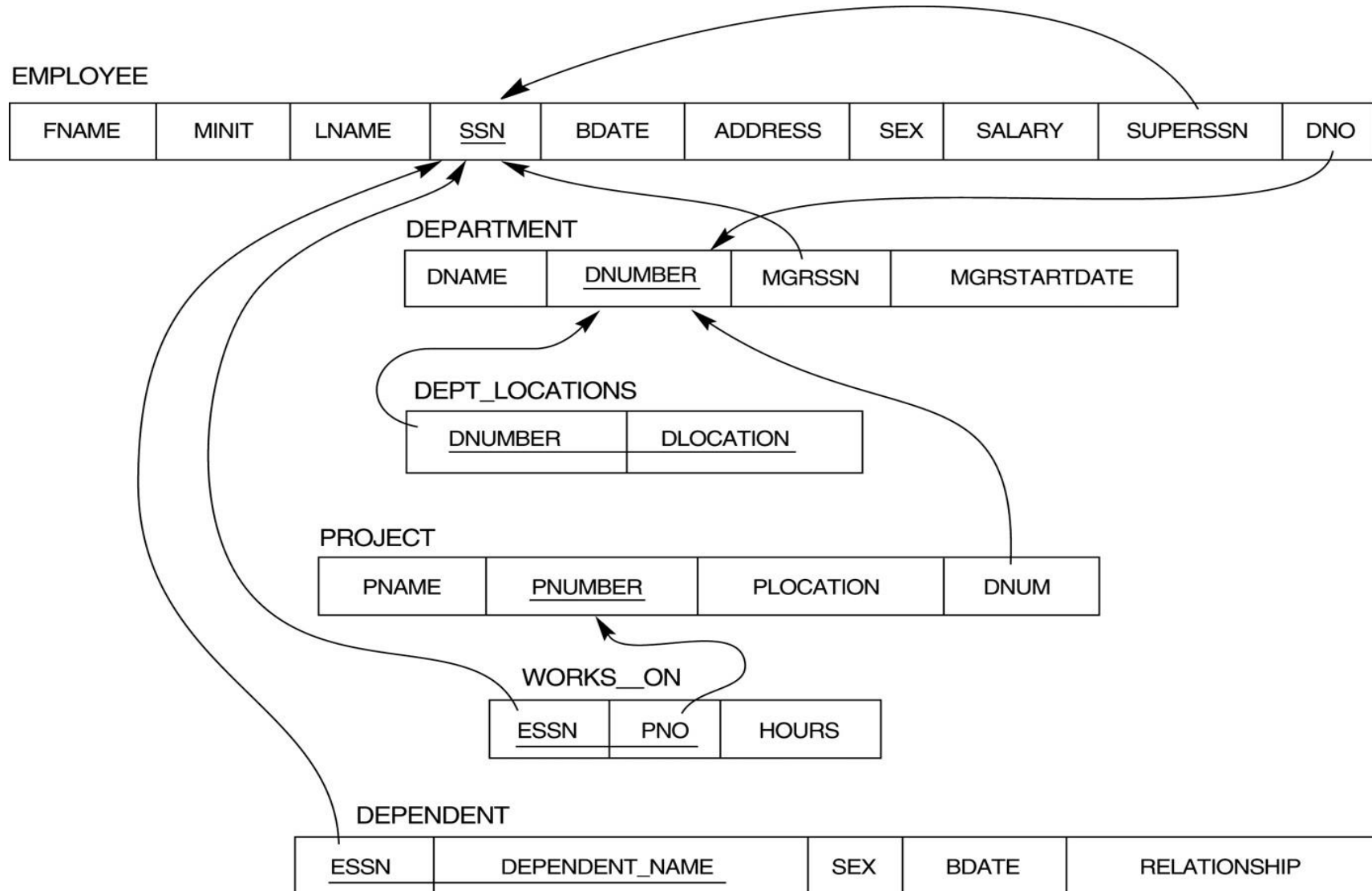
Supply (sname, jname, partNum, quantity, ...)

sname is FK -> Supplier (sname)

jname is FK -> Job (jname)

partNum is FK -> Part (partNum)

Result of mapping the COMPANY - ER schema into a relational schema.



Summary of mapping constructs & constraints

Correspondence between ER and Relational Models

ER Model

Entity type

1:1 or 1:N relationship type

M:N relationship type

n -ary relationship type

Simple attribute

Composite attribute

Multivalued attribute

Relational Model

“Entity” relation

Foreign key (or “relationship” relation)

“Relationship” relation and two foreign keys

“Relationship” relation and n foreign keys

Attribute

Set of simple component attributes

Relation and foreign key

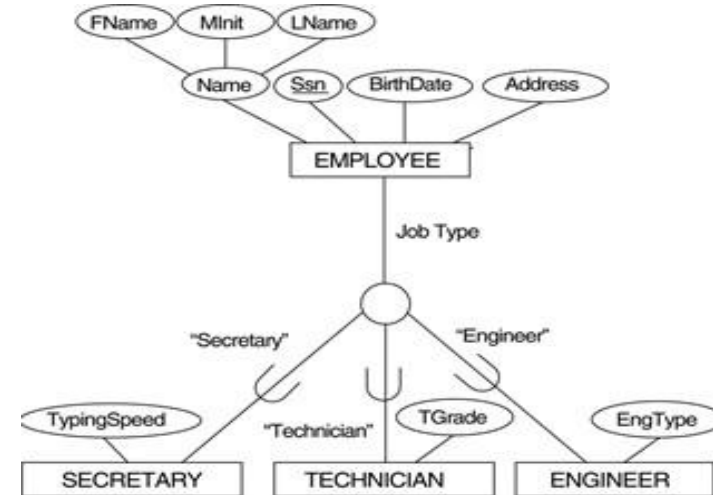
Mapping EER Model Constructs to Relations

- **Step8: Options for Mapping Specialization or Generalization.**
 - Convert each specialization with m subclasses $\{S_1, S_2, \dots, S_m\}$ and generalized superclass C , where the attributes of C are $\{k, a_1, \dots, a_n\}$ and k is the (primary) key, into relational schemas using one of the four following options:
 - **Option 8A: Multiple relations-Superclass and subclasses**
 - **Option 8B: Multiple relations-Subclass relations only**
 - **Option 8C: Single relation with one type attribute**
 - **Option 8D: Single relation with multiple type attributes**

Option 8A

Option 8A: Multiple relations-Superclass and subclasses

Create a relation L for C as $L\{k, a_1, \dots, a_n\}$, and $PK(L) = k$. Create a relation L_i for each subclass S_i , $1 < i < m$, $L_i\{k, s_{i1}, \dots, s_{iik}\}$, where $s_{i1}, \dots, s_{iik}$ are the local attributes of S_i , and $PK(L_i) = k$. This option works for **any specialization** (total or partial, disjoint or over-lapping).



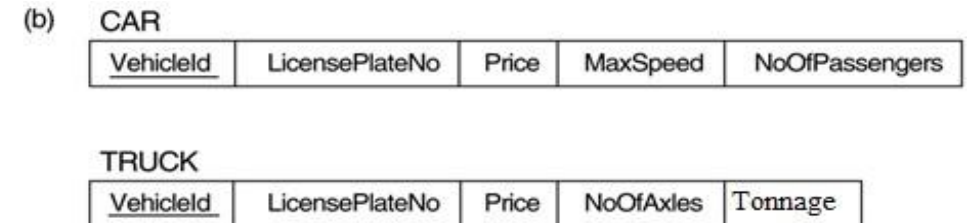
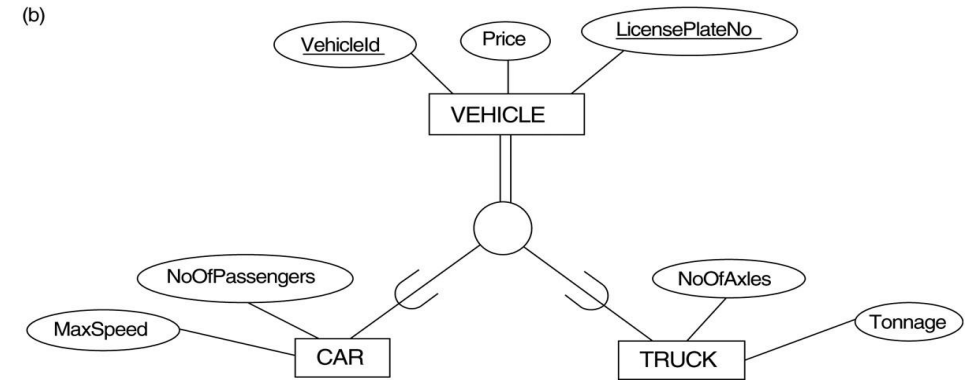
(a)



Option 8B

Option 8B: Multiple relations-Subclass relations only

Create a relation L_i for each subclass S_i , $1 < i < m$, with the attributes $\text{Attr}(L_i) = \{\text{attributes of } S_i\} \cup \{k, a_1, \dots, a_n\}$ and $\text{PK}(L_i) = k$. This option **only** works for a specialization whose subclasses are **total** (every entity in the superclass must belong to (at least) one of the subclasses).

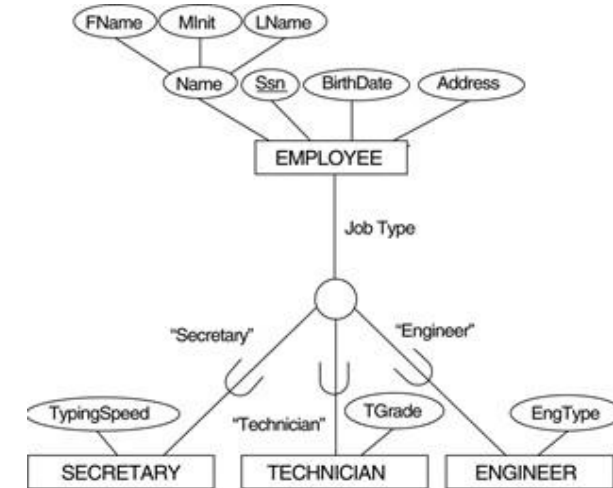


Option 8C

Option 8C: Single relation with one type attribute.

Create a single relation L with attributes $\text{Attrs}(L) = \{k, a_1, \dots, a_n\} \cup \{\text{attributes of } S_1\} \cup \dots \cup \{\text{attributes of } S_m\} \cup \{t\}$ and $\text{PK}(L) = k$. The attribute t is called a type (or **discriminating**) attribute that indicates the subclass to which each tuple belongs.

This option works for **disjoint specilaization**.



(c) EMPLOYEE

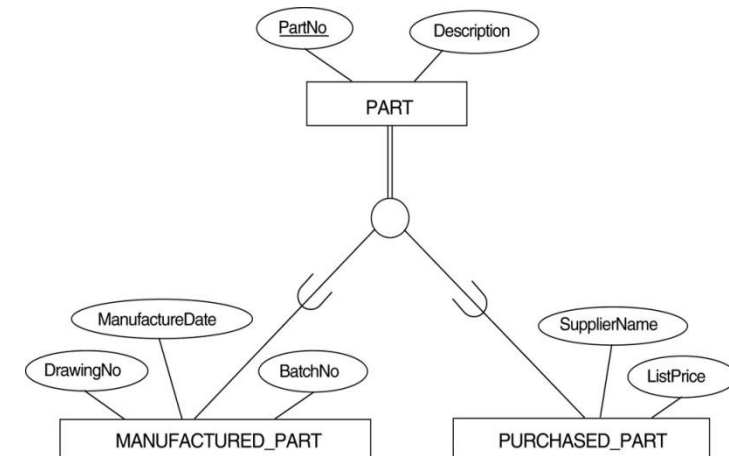
<u>SSN</u>	FName	Minit	LName	BirthDate	Address	JobType	TypingSpeed	TGrade	EngType
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Option 8D

Option 8D: Single relation with multiple type attributes.

Create a single relation schema L with attributes $\text{Attrs}(L) = \{k, a_1, \dots, a_n\} \cup \{\text{attributes of } S_1\} \cup \dots \cup \{\text{attributes of } S_m\} \cup \{t_1, t_2, \dots, t_m\}$ and $\text{PK}(L) = k$. Each t_i , $1 < i < m$, is a Boolean type attribute indicating whether a tuple belongs to the subclass S_i .

This option works for **overlapping specialization**.



(d) PART

<u>PartNo</u>	Description	MFlag	DrawingNo	ManufactureDate	BatchNo	PFlag	SupplierName	ListPrice
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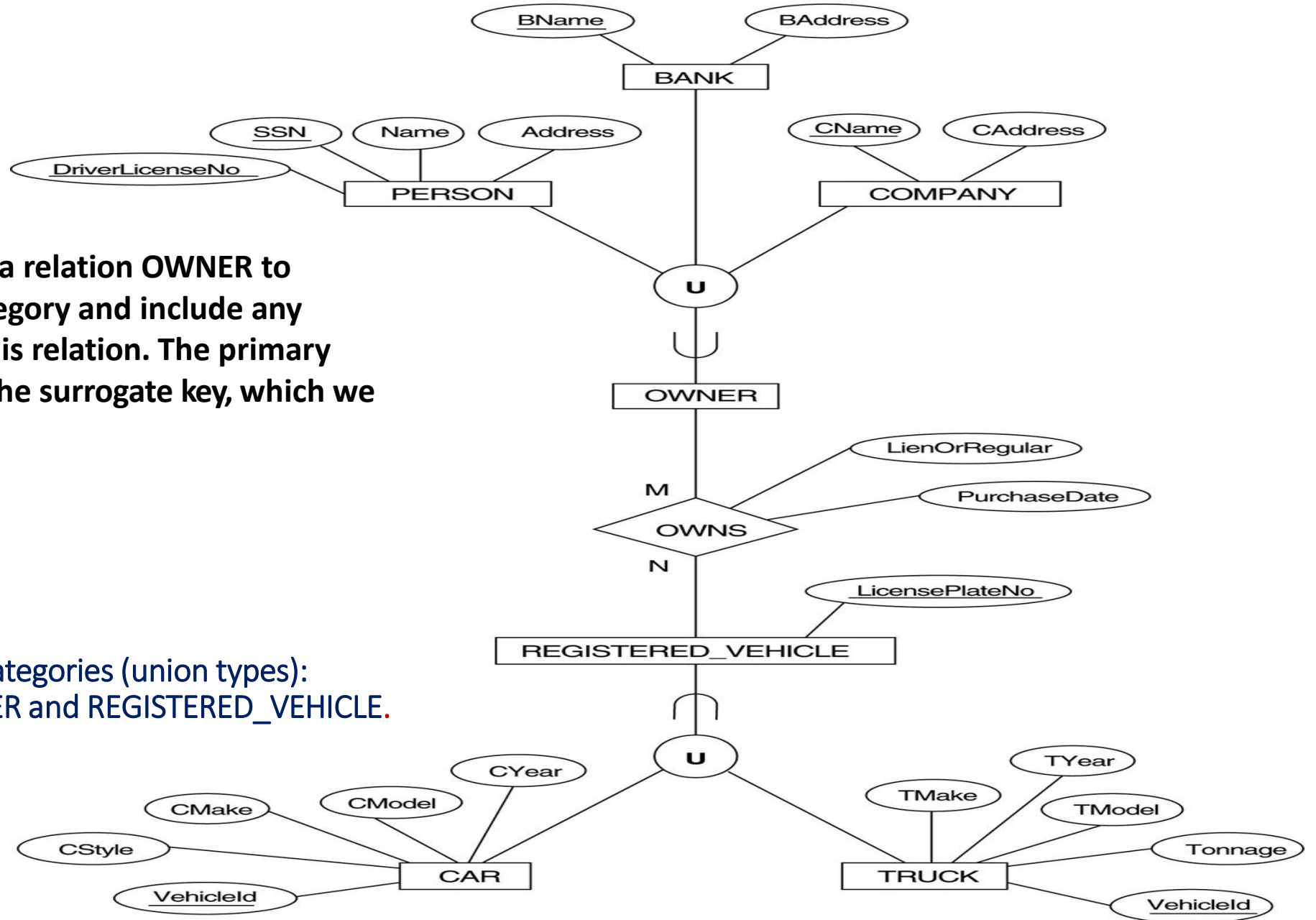
Mapping EER Model Constructs to Relations (cont)

Step 9: Mapping of Union Types (Categories).

- For mapping a category whose defining superclass have different keys, it is customary to specify a new key attribute, called a **surrogate key**, when creating a relation to correspond to the category.

In the **example**, we can create a relation OWNER to correspond to the OWNER category and include any attributes of the category in this relation. The primary key of the OWNER relation is the surrogate key, which we called OwnerId.

Two categories (union types):
OWNER and REGISTERED_VEHICLE.



Mapping the EER categories (union types) to relations.

